Physics Tech 9 — Solved Exercises (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

8 chapters — 286 exercises

Chapter 1: Physical Quantities and Measurements

A. Multiple Choice Questions

1.1 One femtometre is equal to:

(a) 10^-9 m

(b) 10^-15 m ✓

(c) 10^9 m

(d) 10^15 m

Correct answer: 10^-15 m

A. Multiple Choice Questions

1.2 A light year is a unit of:

(a) light

(b) time

(c) distance ✓

(d) speed

Correct answer: distance

A. Multiple Choice Questions

1.3 Which one is a non-physical quantity?

(a) distance

(b) density

(c) colour ✓

(d) temperature

Correct answer: colour

A. Multiple Choice Questions

1.4 Two rods with lengths 12.321 cm and 10.3 cm are placed side by side, the difference in their lengths is:

(a) 2.02 cm

(b) 2.0 cm

(c) 2 cm

(d) 2.021 cm

Correct answer: 2.0 cm

Note on Question 1.4

Note: Why is the difference 2.0 cm and not 2.021 cm?

The raw subtraction gives 12.321 - 10.3 = 2.021 cm. But a result can be no more precise than the least precise measurement used, and 10.3 cm is known to only one decimal place. So the answer is rounded to one decimal place: 2.0 cm. That is why option (b) is correct.

A. Multiple Choice Questions

1.5 Which of the following measures are likely to represent the thickness of a sheet of this book?

(a) 6 x 10^-25 m

(b) 1 x 10^-4 m

(c) 1.2 x 10^-15 m

(d) 4 x 10^-2 m

Correct answer: 1 x 10^-4 m

B. Short Answer Questions

1.1 Can a non-physical quantity be measured? If yes, then how?

No, a non-physical quantity cannot be measured in the scientific sense. Measurement means comparing a quantity with a standard unit and expressing it as a number followed by a unit. Non-physical quantities such as colour, beauty, love or honesty have no standard unit and no measuring instrument, so

they can only be described or compared in words, never given a numerical value with a unit.

B. Short Answer Questions

1.2 What is measurement? Name its two parts.

Measurement is the comparison of an unknown quantity with a known standard quantity of the same kind. Its two parts are: (1) a number, which tells how many times the standard unit is contained in the quantity, and (2) a unit, which is the standard itself, for example 5 m has the number 5 and the unit metre.

B. Short Answer Questions

1.3 Why do we need a standard unit for measurements?

We need a standard unit so that a measurement means the same thing to everyone, everywhere. Without it, people would use hand-spans, foot-steps or arm-lengths, which differ from person to person, so the same object would be reported with different values. A standard unit makes measurements uniform, reproducible and comparable all over the world.

B. Short Answer Questions

1.4 Write the name of 3 base quantities and 3 derived quantities.

Base quantities (independent of all others): length, mass and time. Derived quantities (obtained from base quantities by multiplication or division): speed (length/time), area (length x length) and density (mass/volume). Other examples are force, volume and pressure.

B. Short Answer Questions

1.5 Which SI unit will you use to express the height of your desk?

The metre (m). A desk is roughly a metre high, so the metre is the convenient SI

unit. Millimetres would give a needlessly large number and kilometres would give a tiny fraction.

B. Short Answer Questions

1.6 Write the name and symbols of all SI base units.

Length: metre (m). Mass: kilogram (kg). Time: second (s). Temperature: kelvin (K). Electric current: ampere (A). Amount of substance: mole (mol). Luminous intensity: candela (cd). These seven are the SI base units; every other SI unit is derived from them.

B. Short Answer Questions

1.7 Why prefix is used? Name three sub-multiple and three multiple prefixes with their symbols.

A prefix is used so that very large or very small quantities can be written conveniently, without long strings of zeros. Sub-multiple prefixes (smaller than the unit): milli (m) = 10^-3, micro (u) = 10^-6, nano (n) = 10^-9. Multiple prefixes (larger than the unit): kilo (k) = 10^3, mega (M) = 10^6, giga (G) = 10^9.

B. Short Answer Questions

1.8 What is meant by: (a) 5 pm (b) 15 ns (c) 6 um (d) 5 fs

(a) 5 pm means 5 picometres = 5 x 10^-12 m. (b) 15 ns means 15 nanoseconds = 15 x 10^-9 s. (c) 6 um means 6 micrometres = 6 x 10^-6 m. (d) 5 fs means 5 femtoseconds = 5 x 10^-15 s. In each case the prefix tells the power of ten by which the unit is multiplied.

C. Constructed Response Questions

1.1 In what unit will you express each of the following? (a) Thickness of a

five-rupee coin (b) Length of a book (c) Length of football field (d) The distance between two cities (e) Mass of five-rupee coin (f) Mass of your school bag (g) Duration of your class period (h) Volume of petrol filled in the tank of a car

(a) millimetre (mm), about 2 mm thick. (b) centimetre (cm) or metre (m), about 25 cm. (c) metre (m), about 100 m. (d) kilometre (km). (e) gram (g). (f) kilogram (kg). (g) minute (min). (h) litre (L). In each case the unit is chosen so the number stays convenient, neither huge nor a tiny fraction.

C. Constructed Response Questions

1.2 Why might a standard system of measurement be helpful to a tailor?

A tailor must cut cloth to fit a particular customer, and the customer, the tailor and the cloth merchant may all measure at different times and places. With a standard system, a chest measurement of 90 cm means the same to all of them, so the suit fits whoever sews it. With body-based units such as hand-spans, the same order would come out a different size from every tailor.

C. Constructed Response Questions

1.3 The end of a metre scale is worn out. Where will you place a pencil to find the length?

Place one end of the pencil against a clear, readable mark, for example the 10 cm mark, instead of the worn zero end. Read the position of the other end, say 27 cm, and subtract: length = 27 cm - 10 cm = 17 cm. Subtracting the starting reading removes the error caused by the worn end.

C. Constructed Response Questions

1.4 Why is it better to place the object close to the metre scale?

If the object is held away from the scale, the eye looking at an angle sees the object's end against the wrong mark on the scale. This is parallax error. Placing

the object close to, and touching, the scale keeps the line of sight perpendicular to the mark and gives the true reading.

C. Constructed Response Questions

1.5 Why a standard unit is needed to measure a quantity correctly?

A quantity can only be measured correctly if it is compared with a fixed, agreed standard. If everyone used a different standard, the same quantity would be reported with different numbers and no two measurements could be compared or checked. A standard unit, fixed and the same everywhere, is what makes a measurement correct and meaningful.

C. Constructed Response Questions

1.6 Suggest some natural phenomena that could serve as a reasonably accurate time standard.

The rotation of the Earth on its axis (one day), the revolution of the Earth around the Sun (one year), the revolution of the Moon around the Earth (one month), and the regular swing of a pendulum. In modern clocks the most accurate natural standard is the vibration of atoms, for example caesium atoms, which vibrate at a perfectly steady rate and form the basis of atomic clocks.

D. Comprehensive Questions

1.1 What is meant by base and derived quantities? Give the names and symbols of SI base units.

Base quantities are the independent quantities from which all others are built; they cannot be expressed in terms of one another. They are length, mass, time, temperature, electric current, amount of substance and luminous intensity. Derived quantities are obtained by multiplying or dividing base quantities, for example speed = length/time and force = mass x acceleration. SI base units:

metre (m), kilogram (kg), second (s), kelvin (K), ampere (A), mole (mol) and candela (cd).

D. Comprehensive Questions

1.2 Give three examples of derived unit in SI. How are they derived from base units? Describe briefly.

Speed: its unit is metre per second (m s^-1), derived as length divided by time. Force: its unit is the newton (N), derived as mass x acceleration, that is kg m s^-2. Area: its unit is square metre (m^2), derived as length x length. In each case the derived unit is simply the combination of the base units in the same formula that defines the quantity.

D. Comprehensive Questions

1.3 Identify and explain the reasons for human errors, random errors and systematic errors in experiments.

Human errors are mistakes by the observer, such as misreading a scale, looking at it from an angle (parallax), or recording a wrong digit; they are reduced by care and by repeating the reading. Random errors are unpredictable small variations, for example tiny fluctuations in the apparatus or the surroundings; they make repeated readings differ slightly in either direction and are reduced by taking the average of many readings. Systematic errors shift every reading the same way, for example a scale with a worn zero end or an instrument with a zero error; they are removed by correcting or calibrating the instrument.

D. Comprehensive Questions

1.4 Differentiate between precision and accuracy of a measurement with examples.

Precision is how close repeated measurements of the same thing are to each

other; it depends on the least count of the instrument. Accuracy is how close a measurement is to the true value. Example with a target: shots clustered tightly together but away from the centre are precise but not accurate; shots scattered around the centre are accurate on average but not precise; shots clustered tightly on the centre are both precise and accurate. A measurement can therefore be precise without being accurate, and vice versa.

E. Numerical Problems

1.1 Calculate the number of seconds in a (a) day (b) week (c) month and state your answers using SI prefixes. (Book's printed answers: 86.4 ks, 604.8 ks, 2.592 Ms)

(a) One day: 24 hours x 60 minutes x 60 seconds = 86,400 s. Step 2: 86,400 s = 86.4 x 10^3 s = 86.4 ks. (b) One week: 7 x 86,400 s = 604,800 s = 604.8 ks. (c) One month (30 days): 30 x 86,400 s = 2,592,000 s = 2.592 x 10^6 s = 2.592 Ms. These match the answers printed in the book.

E. Numerical Problems

1.2 State the answers of problem 1.1 in scientific notation. (Book's printed answers: 8.64 x 10^4 s, 6.048 x 10^5 s, 2.592 x 10^6 s)

(a) Day: 86,400 s = 8.64 x 10^4 s. (b) Week: 604,800 s = 6.048 x 10^5 s. (c) Month: 2,592,000 s = 2.592 x 10^6 s. Each number is written with one digit before the decimal point times the correct power of ten, matching the book's printed answers.

E. Numerical Problems

1.3 Solve the following addition or subtraction. State your answers in scientific notation. (a) 4 x 10^-4 kg + 3 x 10^-5 kg (b) 5.4 x 10^-6 m - 3.2 x 10^-5 m. (Book's printed answers: (a) 4.3 x 10^-4 kg, (b) -2.66 x 10^-5 m)

(a) Step 1: express both in the same power: 3 x 10^-5 kg = 0.3 x 10^-4 kg. Step 2: add: 4 x 10^-4 + 0.3 x 10^-4 = 4.3 x 10^-4 kg. (b) Step 1: 5.4 x 10^-6 m = 0.54 x 10^-5 m. Step 2: subtract: 0.54 x 10^-5 - 3.2 x 10^-5 = -2.66 x 10^-5 m. Both match the book's printed answers.

E. Numerical Problems

1.4 Solve the following multiplication or division. State your answers in scientific notation. (a) (5 x 10^4 m) x (3 x 10^-2 m) (b) (6 x 10^8 kg) / (3 x 10^4 m^3). (Book's printed answers: (a) 1.5 x 10^3 m^2, (b) 2.0 x 10^4 kg m^-3)

(a) Step 1: multiply the numbers: 5 x 3 = 15. Step 2: add the powers: 10^4 x 10^-2 = 10^2. Step 3: 15 x 10^2 = 1.5 x 10^3 m^2. (b) Step 1: divide the numbers: 6/3 = 2. Step 2: subtract the powers: 10^8 / 10^4 = 10^4. Step 3: 2.0 x 10^4 kg m^-3. Both match the book's printed answers.

E. Numerical Problems

1.5 Calculate the following and state your answer in scientific notation. [(3 x 10^2 kg) x (4.0 km)] / (5 x 10^2 s^2). (Book's printed answer: 2.4 x 10^3 kg m s^-2)

Step 1: convert km to m: 4.0 km = 4.0 x 10^3 m. Step 2: multiply the numerator: 3 x 10^2 x 4.0 x 10^3 = 12 x 10^5 = 1.2 x 10^6 kg m. Step 3: divide by the denominator: 1.2 x 10^6 / 5 x 10^2 = 0.24 x 10^4 = 2.4 x 10^3 kg m s^-2. This matches the book's printed answer.

E. Numerical Problems

1.6 State the number of significant digits in each measurement. (a) 0.0045 m (b) 2.047 m (c) 3.40 m (d) 3.420 x 10^4 m. (Book's printed answers: (a) 2, (b) 4, (c) 3, (d) 4)

(a) 0.0045 m: leading zeros are not significant, so 4 and 5 count: 2 significant

digits. (b) 2.047 m: zeros between digits are significant: 4 significant digits. (c) 3.40 m: the trailing zero after the decimal point is significant: 3 significant digits. (d) 3.420 x 10^4 m: all four digits in the coefficient count: 4 significant digits. All match the book's printed answers.

E. Numerical Problems

1.7 Write in scientific notation: (a) 0.0035 m (b) 206.4 x 10^2 m. (Book's printed answers: (a) 3.5 x 10^-3 m, (b) 2.064 x 10^4 m)

(a) Move the decimal point three places right: 0.0035 = 3.5 x 10^-3 m. (b) First write 206.4 as 2.064 x 10^2, then 2.064 x 10^2 x 10^2 = 2.064 x 10^4 m. Both match the book's printed answers.

E. Numerical Problems

1.8 Write using correct prefixes: (a) 5.0 x 10^4 cm (b) 580 x 10^2 g (c) 45 x 10^-4 s. (Book's printed answers: 0.5 km, 58 kg, 4.5 ms)

(a) Step 1: 5.0 x 10^4 cm = 5.0 x 10^2 m = 500 m. Step 2: 500 m = 0.5 km. (b) Step 1: 580 x 10^2 g = 58,000 g. Step 2: 58,000 g = 58 kg. (c) Step 1: 45 x 10^-4 s = 4.5 x 10^-3 s. Step 2: 10^-3 s is a millisecond, so 4.5 ms. All match the book's printed answers.

Chapter 2: Kinematics

A. Multiple Choice Questions

2.1 The numerical ratio of displacement to distance is:

(a) always less than one

(b) always equal to one

(c) always greater than one

(d) equal to or less than one ✓

Correct answer: equal to or less than one

A. Multiple Choice Questions

2.2 A ball is dropped from the top of a tower, the distance covered by it in the first second is:

(a) 5 m ✓

(b) 10 m

(c) 50 m

(d) 100 m

Correct answer: 5 m

Working for Question 2.2

Working: distance in the first second of free fall

For a freely falling body, initial velocity vi = 0 and acceleration g = 10 m s^-2. Distance S = vi x t + (1/2) g t^2 = 0 + (1/2) x 10 x (1)^2 = 5 m. So option (a) 5 m is correct.

A. Multiple Choice Questions

2.3 A body accelerates from rest to a velocity of 144 km h^-1 in 20 seconds. The distance covered by it is:

(a) 100 m

(b) 400 m

(c) 1400 m

(d) 1440 m

Correct answer: 400 m

Working for Question 2.3

Working: distance from rest to 144 km h^-1 in 20 s

Step 1: convert velocity: 144 km h^-1 = 144 x 1000 / 3600 = 40 m s^-1. Step 2: acceleration a = (40 - 0) / 20 = 2 m s^-2. Step 3: distance S = (1/2) a t^2 = (1/2) x 2 x (20)^2 = 400 m. So option (b) 400 m is correct.

A. Multiple Choice Questions

2.4 A body is moving with constant acceleration starting from rest. It covers a distance S in 4 seconds. How much time does it take to cover one-fourth of this distance?

(a) 1 s

(b) 2 s

(c) 4 s

(d) 16 s

Correct answer: 2 s

Working for Question 2.4

Working: time for one-fourth distance

From rest, distance S = (1/2) a t^2, so distance is proportional to the square of time. One-fourth of the distance needs the square root of one-fourth of the time squared: t_new = 4 s x sqrt(1/4) = 4 x (1/2) = 2 s. So option (b) 2 s is correct.

A. Multiple Choice Questions

2.5 The displacement-time graphs of two objects A and B are shown in the figure. Point out the true statement from the following:

(a) The velocity of A is greater than B. ✓

(b) The velocity of A is less than B.

(c) The velocity of A is equal to that of B.

(d) The graph gives no information in this regard.

Correct answer: The velocity of A is greater than B.

Working for Question 2.5

Working: why is the velocity of A greater?

Velocity is the gradient (slope) of the displacement-time graph. Graph A is a steep rising line, so it has a large positive gradient. Graph B is a horizontal line, so its gradient, and therefore its velocity, is zero. Since A's gradient is larger, the velocity of A is greater than B. So option (a) is correct.

A. Multiple Choice Questions

2.6 Gradient of the speed-time graph is equal to:

(a) speed

(b) velocity

(c) acceleration ✓

(d) distance covered

Correct answer: acceleration

A. Multiple Choice Questions

2.7 Gradient of the distance-time graph is equal to the:

(a) speed ✓

(b) velocity

(c) distance covered

(d) acceleration

Correct answer: speed

Note on Question 2.7

Note: speed or velocity?

The gradient of a distance-time graph is speed, because distance is a scalar. The gradient of a displacement-time graph is velocity. For motion in a straight line without turning back, the numerical values are the same, which is why the two words are sometimes mixed up, but for the distance-time graph as printed, option (a) speed is the exact answer.

A. Multiple Choice Questions

2.8 A car accelerates uniformly from 80.5 km h^-1 at t = 0 to 113 km h^-1 at t = 9 s. Which graph best describes the motion of the car?

(a) Graph (a): curved line rising from a positive value

(b) Graph (b): straight line rising from zero

(c) Graph (c): straight line rising from a positive value

(d) Graph (d): straight line falling to zero

Correct answer: Graph (c): straight line rising from a positive value

Working for Question 2.8

Working: which velocity-time graph fits?

Uniform acceleration means the velocity-time graph is a straight line, so the curved graph (a) is wrong. The car already moves at 80.5 km h^-1 at t = 0, so the line must start above zero, which rules out graph (b) starting at zero. The velocity increases to 113 km h^-1, so the line must rise, ruling out falling graph (d). Only graph (c), a straight rising line starting from a positive velocity, fits.

B. Short Answer Questions

2.1 Define scalar and vector quantities.

A scalar quantity has magnitude only, with no direction, for example mass, time, distance and temperature. A vector quantity has both magnitude and direction, for example displacement, velocity, acceleration and force. A vector must therefore always be stated with its direction as well as its size.

B. Short Answer Questions

2.2 Give 5 examples each for scalar and vector quantities.

Scalars: mass, time, distance, speed and temperature (also energy and volume). Vectors: displacement, velocity, acceleration, force and momentum (also weight and torque). The test is simple: if the quantity needs a direction to be fully described, it is a vector; otherwise it is a scalar.

B. Short Answer Questions

2.3 State head-to-tail rule for addition of vectors.

To add two vectors, draw the first vector, then draw the second vector with its tail at the head of the first. The resultant vector is drawn from the tail of the first vector to the head of the second vector. Its length gives the magnitude of the resultant and its arrowhead gives the direction.

B. Short Answer Questions

2.4 What are distance-time graph and speed-time graph?

A distance-time graph plots distance travelled against time; its gradient at any point gives the speed, and a steeper line means a higher speed. A speed-time graph plots speed against time; its gradient gives the acceleration, and the area under the graph gives the distance covered.

B. Short Answer Questions

2.5 Falling objects near the Earth have the same constant acceleration. Does this imply that a heavier object will fall faster than a lighter object?

No. Near the Earth every freely falling object has the same acceleration, g = 10 m s^-2, whatever its mass, because the larger gravitational force on a heavier object is exactly balanced by its larger inertia. In air, a light object with a large surface may fall more slowly because of air resistance, but in the absence of air resistance both fall together.

B. Short Answer Questions

2.6 The vector quantities are sometimes written in scalar notation (not bold face). How is the direction indicated?

The direction is then shown by an arrow drawn over the symbol, by a plus or minus sign for motion along a straight line (for example + for one direction and - for the opposite), or by stating the angle the vector makes with a fixed reference direction such as the x-axis.

B. Short Answer Questions

2.7 A body is moving with uniform speed. Will its velocity be uniform? Give reason.

Not necessarily. Velocity is uniform only if both speed and direction stay constant. A body moving in a circle at uniform speed keeps changing direction, so its velocity keeps changing even though its speed is constant. Uniform speed therefore guarantees uniform velocity only when the body also moves in a straight line.

B. Short Answer Questions

2.8 Is it possible for a body to have acceleration? When moving with: (i) constant

velocity (ii) constant speed

(i) No. Constant velocity means neither speed nor direction changes, so the change in velocity is zero and acceleration is zero. (ii) Yes. Constant speed with changing direction, as in uniform circular motion, still changes the velocity, so the body has acceleration directed towards the centre of the circle.

C. Constructed Response Questions

2.1 Distance and displacement may or may not be equal in magnitude. Explain this statement.

Distance is the total length of the path actually travelled, while displacement is the straight-line distance from the starting point to the finishing point, with direction. If a body moves in a straight line without turning back, the path length equals the straight-line distance, so they are equal. If it turns back or follows a curved path, the path travelled is longer than the straight-line change of position, so distance is greater than the magnitude of displacement. Distance can never be less than displacement.

C. Constructed Response Questions

2.2 For a complete trip, average velocity was calculated. Its value came out to be positive. Is it possible that its instantaneous velocity at any time during the trip had the negative value? Give justification of your answer.

Yes, it is possible. Average velocity depends only on the total displacement divided by the total time, so a positive average means only that the finishing point lies in the positive direction from the starting point. During the trip the body may have moved backwards for a while, giving negative instantaneous velocity at those times, as long as its forward motion over the whole trip was greater. For example, a car that drives forward, reverses briefly into a parking space, then drives on forward still has a positive average velocity.

C. Constructed Response Questions

2.3 A ball is thrown vertically upward with velocity v. It returns to the ground in time T. Which of the following graphs correctly represents the motion? Explain your reasoning.

Graph (b) is correct. The ball starts with velocity +v upward. Gravity reduces its velocity steadily, so the velocity-time graph is a straight line sloping downwards. At half the total time, T/2, the ball is momentarily at rest at its highest point, so the line crosses zero at T/2. It then falls back with increasing downward velocity and reaches the ground at time T with velocity -v. Graph (a) is a V-shape, which shows speed, not velocity, because it never goes negative. Graph (c) crosses zero after T/2, so its timing is wrong. Graph (d) starts at zero and is never negative, so it cannot describe a ball that returns to the ground.

C. Constructed Response Questions

2.4 The figure given below shows the distance-time graph for the travel of a cyclist. Find the velocities for the segments a, b and c.

Velocity is the gradient of the distance-time graph. Segment a: the cyclist goes from 0 to 2.0 km in 6 minutes. Velocity = 2.0 km / 6 min = 0.333 km/min = 20 km h^-1 away from the start. Segment b: the distance stays at 2.0 km from 6 to 11 minutes, so the change in distance is zero and the velocity is 0; the cyclist is at rest. Segment c: the cyclist returns from 2.0 km to 0 in 9 minutes (from 11 to 20 minutes). Velocity = -2.0 km / 9 min = -0.222 km/min = -13.3 km h^-1; the negative sign means motion back towards the starting point at a speed of 13.3 km h^-1.

C. Constructed Response Questions

2.5 Is it possible that the velocity of an object is zero at an instant of time, but its acceleration is not zero? If yes, give an example of such a case.

Yes. Acceleration is the rate of change of velocity, not the velocity itself, so a body can be momentarily at rest while its velocity is still changing. The classic example is a ball thrown vertically upward: at the very top of its flight its velocity is zero for an instant, but gravity still acts on it, so its acceleration is g = 10 m s^-2 downwards, and in the next instant it begins to fall.

D. Comprehensive Questions

2.1 How a vector can be represented graphically? Explain.

A vector is represented graphically by an arrow. The length of the arrow is drawn proportional to the magnitude of the vector, using a chosen scale such as 1 cm = 10 N. The direction in which the arrow points is the direction of the vector, and the arrowhead shows which end is the head. The tail of the arrow marks the point where the vector acts. Two vectors are equal if their arrows have the same length and point in the same direction.

D. Comprehensive Questions

2.2 Differentiate between: (i) rest and motion (ii) speed and velocity

(i) Rest and motion: a body is at rest if its position does not change with time relative to its surroundings; it is in motion if its position changes with time. Rest and motion are relative: a passenger sitting in a moving bus is at rest relative to the bus but in motion relative to the road. (ii) Speed and velocity: speed is a scalar giving only how fast a body moves, for example 60 km h^-1. Velocity is a vector giving how fast and in which direction it moves, for example 60 km h^-1 towards the north. Speed is the magnitude of velocity.

D. Comprehensive Questions

2.3 What do gradients of distance-time graph and speed-time graph represent? Explain it by drawing diagrams.

The gradient of a distance-time graph represents speed: a steeper line means a greater speed, a horizontal line means rest, and a line sloping downwards means the body is returning. The gradient of a speed-time graph represents acceleration: a rising line means acceleration, a falling line means deceleration, and a horizontal line means uniform speed. Diagrams to draw: (1) a distance-time graph with a straight rising line labelled with its gradient = distance/time = speed; (2) a speed-time graph with a straight rising line labelled with its gradient = change in speed/time = acceleration. Both diagrams are straight-line graphs with labelled axes, gradient triangles drawn on them, and the gradient values written beside the triangles.

D. Comprehensive Questions

2.4 Prove that the area under speed-time graph is equal to the distance covered by an object.

For uniform speed, speed v = distance/time, so distance = speed x time. On a speed-time graph, speed x time is the area of the rectangle under the horizontal line, so the area equals the distance. For changing speed, divide the time into many tiny intervals; in each tiny interval the speed is nearly constant, so the distance in that interval is speed x tiny time, which is the area of a thin strip under the graph. Adding all the strips gives the total area under the graph, which is therefore the total distance covered.

D. Comprehensive Questions

2.5 How equations of motion can be applied to the bodies moving under the action of gravity?

Bodies moving under gravity use the same three equations of motion, with acceleration a replaced by the acceleration due to gravity, g = 10 m s^-2, and distance S replaced by height h. Choose a sign convention first: if upward is

taken as positive, then a = -g because gravity acts downwards. The equations become: vf = vi - g t; h = vi t - (1/2) g t^2; and vf^2 = vi^2 - 2 g h. For a freely falling body dropped from rest, vi = 0, so they simplify to vf = g t, h = (1/2) g t^2 and vf^2 = 2 g h. The same equations give the time to reach the highest point (where vf = 0) and the velocity on return to the ground.

E. Numerical Problems

2.1 Draw the representative lines of the following vectors: (a) A velocity of 400 m s^-1 making an angle of 60 degrees with x-axis. (b) A force of 50 N making an angle of 120 degrees with x-axis.

Drawing task, described so it can be drawn exactly: (a) Draw the x-axis. From the origin draw an arrow 4 cm long if the scale is 1 cm = 100 m s^-1, pointing up and to the right so that it makes 60 degrees with the positive x-axis; label it 400 m s^-1. (b) From the same origin draw an arrow 5 cm long if the scale is 1 cm = 10 N, pointing up and to the left so that it makes 120 degrees with the positive x-axis, that is 60 degrees above the negative x-axis; label it 50 N. In both cases the arrow length shows the magnitude and the arrowhead shows the direction.

E. Numerical Problems

2.2 A car is moving with an average speed of 72 km h^-1. How much time will it take to cover a distance of 360 km? (Book's printed answer: 5 h)

Step 1: write the data: average speed v = 72 km h^-1, distance S = 360 km, time t = ? Step 2: use v = S/t, so t = S/v. Step 3: t = 360/72 = 5 h. This matches the book's printed answer.

E. Numerical Problems

2.3 A truck starts from rest. It reaches a velocity of 90 km h^-1 in 50 seconds. Find its average acceleration. (Book's printed answer: 0.5 m s^-2)

Step 1: convert velocity: 90 km h^-1 = 90 x 1000/3600 = 25 m s^-1. Step 2: initial velocity vi = 0, final velocity vf = 25 m s^-1, time t = 50 s. Step 3: average acceleration a = (vf - vi)/t = 25/50 = 0.5 m s^-2. This matches the book's printed answer.

E. Numerical Problems

2.4 A motorcycle initially travelling at 18 km h^-1 accelerates at constant rate of 2 m s^-2. How far will the motorcycle go in 10 seconds? (Book's printed answer: 150 m)

Step 1: convert initial velocity: 18 km h^-1 = 18 x 1000/3600 = 5 m s^-1. Step 2: use S = vi t + (1/2) a t^2. Step 3: S = 5 x 10 + (1/2) x 2 x (10)^2 = 50 + 100 = 150 m. This matches the book's printed answer.

E. Numerical Problems

2.5 A wagon is moving on the road with a velocity of 54 km h^-1. Brakes are applied suddenly. The wagon covers a distance of 25 m before stopping. Determine the acceleration of the wagon. (Book's printed answer: -4.5 m s^-2)

Step 1: convert velocity: 54 km h^-1 = 54 x 1000/3600 = 15 m s^-1. Step 2: final velocity vf = 0, distance S = 25 m. Step 3: use vf^2 = vi^2 + 2 a S, so 0 = (15)^2 + 2 x a x 25. Step 4: 0 = 225 + 50 a, so a = -225/50 = -4.5 m s^-2. The negative sign shows deceleration. This matches the book's printed answer.

E. Numerical Problems

2.6 A ball is dropped from the top of a tower. The ball reaches the ground in 5 seconds. Find the height of the tower and the velocity of the ball with which it strikes the ground. (Book's printed answers: 125 m, 50 m s^-1)

Step 1: vi = 0, g = 10 m s^-2, t = 5 s. Step 2: height h = (1/2) g t^2 = (1/2) x 10 x (5)^2 = 5 x 25 = 125 m. Step 3: striking velocity vf = g t = 10 x 5 = 50 m

s^-1. Both match the book's printed answers.

E. Numerical Problems

2.7 A cricket ball is hit so that it travels straight up in the air. An observer notes that it took 3 seconds to reach the highest point. What was the initial velocity of the ball? If the ball was hit 1 m above the ground, how high did it rise from the ground? (Book's printed answers: 30 m s^-1, 46 m)

Step 1: at the highest point vf = 0, time t = 3 s, g = 10 m s^-2. Step 2: vf = vi - g t, so 0 = vi - 10 x 3, giving vi = 30 m s^-1. Step 3: height risen above the hit point h = vi t - (1/2) g t^2 = 30 x 3 - (1/2) x 10 x 9 = 90 - 45 = 45 m. Step 4: height above the ground = 45 + 1 = 46 m. Both match the book's printed answers.

Chapter 3: Dynamics

A. Multiple Choice Questions

3.1 When we kick a stone, we get hurt. This is due to:

(a) inertia

(b) velocity

(c) momentum

(d) reaction ✓

Correct answer: reaction

A. Multiple Choice Questions

3.2 An object will continue its motion with constant acceleration until:

(a) the resultant force on it begins to decrease.

(b) the resultant force on it is zero. ✓

(c) the resultant force on it begins to increase.

(d) the resultant force is at right angle to its tangential velocity.

Correct answer: the resultant force on it is zero.

Working for Question 3.2

Working: why does constant acceleration stop when the force becomes zero?

By Newton's second law, acceleration a = F/m, so constant acceleration needs a constant resultant force. The moment the resultant force becomes zero, the acceleration also becomes zero and the body continues with whatever velocity it has reached, moving uniformly. So option (b) is correct.

A. Multiple Choice Questions

3.3 A ball with initial momentum p hits a solid wall and bounces back with the same velocity. Its momentum p' after collision will be:

(a) p' = p

(b) p' = -p

(c) p' = 2p

(d) p' = -2p

Correct answer: p' = -p

A. Multiple Choice Questions

3.4 A particle of mass m moving with a velocity v collides with another particle of the same mass at rest. The velocity of the first particle after collision is:

(a) v

(b) -v

(c) 0

(d) -1/2

Correct answer: 0

Working for Question 3.4

Working: equal-mass elastic collision

In an elastic collision of two equal masses, the moving particle stops and the stationary particle moves off with the first particle's velocity; momentum and kinetic energy are both conserved only in this exchange. The first particle's velocity after collision is therefore 0, option (c).

A. Multiple Choice Questions

3.5 Conservation of linear momentum is equivalent to:

(a) Newton's first law of motion

(b) Newton's second law of motion

(c) Newton's third law of motion

(d) None of these

Correct answer: Newton's third law of motion

Note on Question 3.5

Note: why is conservation of momentum linked to the third law?

In a collision, the action and reaction forces are equal and opposite and act for the same time, so the momentum gained by one body exactly equals the momentum lost by the other. The total therefore stays constant. That is why conservation of linear momentum is presented as equivalent to Newton's third law, option (c).

A. Multiple Choice Questions

3.6 A large force acts on an object for a very short interval of time. In this case, it is easy to determine:

(a) magnitude of force

(b) time interval

(c) product of force and time ✓

(d) none of these

Correct answer: product of force and time

A. Multiple Choice Questions

3.7 A lubricant is usually introduced between two surfaces to decrease friction. The lubricant:

(a) decreases temperature

(b) acts as ball bearings

(c) prevents direct contact of the surfaces ✓

(d) provides rolling friction

Correct answer: prevents direct contact of the surfaces

B. Short Answer Questions

3.1 What kind of changes in motion may be produced by a force?

A force can start a body moving, stop a moving body, change its speed, change its direction, or change its shape or size. In short, a force changes the state of motion or the shape of a body.

B. Short Answer Questions

3.2 An object moves with constant velocity in free space. How long will the object continue to move with this velocity?

It will continue forever, unless an external force acts on it. This is Newton's first law: in free space there is no friction or other force to slow it down, so a body in uniform motion keeps its velocity indefinitely.

B. Short Answer Questions

3.3 Define impulse of force.

Impulse is the product of a force and the time for which it acts: impulse = F x t. It equals the change in momentum produced by the force, and its unit is newton-second (N s).

B. Short Answer Questions

3.4 When sitting in a car which suddenly accelerates from rest, you are pushed back into the seat, why?

Because of inertia. Your body tends to remain at rest while the car accelerates forward beneath you, so relative to the car you are pressed back into the seat until the seat pushes you forward and gives you the car's acceleration.

B. Short Answer Questions

3.5 The force expressed in Newton's second law is a net force. Why is it so?

Because acceleration is produced only by the unbalanced, resultant force. Several forces may act on a body at once, and if they cancel each other the body

does not accelerate at all. Newton's second law, F = m a, therefore uses the vector sum of all the forces, the net force, not any single force on its own.

B. Short Answer Questions

3.6 How can you show that rolling friction is lesser than the sliding friction?

Take a heavy block and pull it across a rough table with a spring balance; note the large force needed to slide it. Now put the same block on rollers or on a small trolley with wheels and pull it again; the spring balance shows a much smaller force. The smaller pulling force for the same load shows that rolling friction is much less than sliding friction.

B. Short Answer Questions

3.7 Define terminal velocity of an object.

Terminal velocity is the constant maximum velocity reached by a falling object when the upward forces on it, air resistance and upthrust, exactly balance its downward weight. The net force then becomes zero, so the object stops accelerating and falls at a steady velocity.

B. Short Answer Questions

3.8 An astronaut walking in space wants to return to his spaceship by firing a hand rocket. In what direction does he fire the rocket?

He fires the rocket away from the spaceship, that is in the direction opposite to the one in which he wants to travel. The exhaust gases go one way and the reaction pushes the astronaut the opposite way, towards the spaceship, by Newton's third law.

C. Constructed Response Questions

3.1 Inflatable airbags are installed in the vehicles as safety equipment. In terms

of momentum, what is the advantage of airbags over seatbelts?

In a crash the driver's momentum must fall to zero, and the force felt is the change in momentum divided by the time taken: F = change in momentum / time. An airbag increases the stopping time much more than a seatbelt alone and also spreads the force over a larger area of the body, so the force on the driver is much smaller and injuries are reduced.

C. Constructed Response Questions

3.2 When a cricket ball hits high, a fielder tries to catch it. While holding the ball he/she draws hands backward. Why?

Drawing the hands backward increases the time over which the ball's momentum falls to zero. Since force = change in momentum / time, a longer time means a smaller force on the hands, so the catch does not sting or injure the fielder.

C. Constructed Response Questions

3.3 When someone jumps from a small boat onto the river bank, why does the jumper often fall into the water? Explain.

By Newton's third law, as the jumper pushes the boat backward to jump forward, the boat pushes the jumper forward with an equal force. But the boat is light and free to move, so it moves backward away from the bank. The jumper's support therefore slips away beneath him, and if he does not jump far enough he lands in the water between the boat and the bank.

C. Constructed Response Questions

3.4 Imagine that if friction vanishes suddenly from everything, then what could be the scenario of daily life activities?

Life would become almost impossible. We could not walk, because walking

needs friction between our feet and the ground. Vehicles could neither start nor stop, and would skid uncontrollably. We could not hold anything, write with a pencil, or tie knots, because all of these depend on friction. Objects placed on tables would slide off at the slightest tilt, and even screws and nails would slip out.

D. Comprehensive Questions

3.1 Describe Newton's laws of motion.

First law (law of inertia): a body stays at rest or in uniform motion in a straight line unless an unbalanced force acts on it. Second law: the acceleration of a body is directly proportional to the net force on it and inversely proportional to its mass, F = m a, in the direction of the force. Third law: to every action there is an equal and opposite reaction; forces always occur in pairs acting on different bodies.

D. Comprehensive Questions

3.2 Define momentum and express Newton's 2nd law of motion in terms of change in momentum.

Momentum is the product of mass and velocity: p = m v, with unit kg m s^-1. Newton's second law in terms of momentum states that the net force on a body equals the rate of change of its momentum: F = change in momentum / time = (m vf - m vi) / t. When mass is constant this gives F = m (vf - vi) / t = m a, the familiar form of the second law.

D. Comprehensive Questions

3.3 State and explain the principle of conservation of momentum.

The principle states that the total momentum of an isolated system stays constant if no external force acts on it. Explanation: in a collision the two bodies

exert equal and opposite forces on each other for the same time, so one gains exactly as much momentum as the other loses. Examples are a gun recoiling when a bullet is fired, and two trolleys pushing apart: the total momentum before the event equals the total momentum after it.

D. Comprehensive Questions

3.4 Describe the motion of a block on a table taking into account the friction between the two surfaces. What is the static friction and kinetic friction?

When a small force is applied to a block on a table, static friction grows to match it and the block stays at rest; this self-adjusting friction is static friction. Its maximum value is the limiting friction; once the applied force exceeds it, the block starts to slide. While sliding, the friction is kinetic friction, which is slightly smaller than limiting static friction and roughly constant, opposing the motion. That is why it is harder to start a block moving than to keep it moving.

D. Comprehensive Questions

3.5 Explain the effect of friction on the motion of vehicles in context of tyre surface and braking force.

Friction between the tyres and the road is what lets a vehicle start, turn and stop: the engine drives the wheels, and friction pushes the vehicle forward, while braking friction slows it. A tyre with a good tread grips the road and gives a large braking force, so the stopping distance is short. A smooth or worn tyre, or a wet or icy road, reduces friction, so the braking force is smaller and the vehicle takes much longer to stop and may skid. Friction is therefore essential for control, but it also wastes energy as heat and wears the tyres.

E. Numerical Problems

3.1 A 10 kg block is placed on a smooth horizontal surface. A horizontal force of

5 N is applied to the block. Find: (a) the acceleration produced in the block. (b) the velocity of block after 5 seconds. (Book's printed answers: 0.5 m s^-2, 2.5 m s^-1)

(a) Step 1: use F = m a, so a = F/m. Step 2: a = 5/10 = 0.5 m s^-2. (b) Step 1: initial velocity vi = 0, time t = 5 s. Step 2: v = vi + a t = 0 + 0.5 x 5 = 2.5 m s^-1. Both match the book's printed answers.

E. Numerical Problems

3.2 The mass of a person is 80 kg. What will be his weight on the Earth? What will be his weight on the Moon? The value of acceleration due to gravity of Moon is 1.6 m s^-2. (Book's printed answers: 800 N, 128 N)

Step 1: weight W = m g. Step 2: on Earth, g = 10 m s^-2, so W = 80 x 10 = 800 N. Step 3: on the Moon, g = 1.6 m s^-2, so W = 80 x 1.6 = 128 N. Both match the book's printed answers.

E. Numerical Problems

3.3 What force is required to increase the velocity of 800 kg car from 10 m s^-1 to 30 m s^-1 in 10 seconds? (Book's printed answer: 1600 N)

Step 1: acceleration a = (vf - vi)/t = (30 - 10)/10 = 2 m s^-2. Step 2: force F = m a = 800 x 2 = 1600 N. This matches the book's printed answer.

E. Numerical Problems

3.4 A 5 g bullet is fired by a gun. The bullet moves with a velocity of 300 m s^-1. If the mass of the gun is 10 kg, find the recoil speed of the gun. (Book's printed answer: -0.15 m s^-1)

Step 1: convert bullet mass: 5 g = 0.005 kg. Step 2: total momentum before firing is zero. Step 3: by conservation of momentum, 0 = 0.005 x 300 + 10 x v. Step 4: 10 v = -1.5, so v = -0.15 m s^-1. The negative sign means the gun

recoils in the opposite direction to the bullet. This matches the book's printed answer.

E. Numerical Problems

3.5 An astronaut weighs 70 kg. He throws a wrench of mass 300 g at a speed of 3.5 m s^-1. Determine: (a) the speed of astronaut as he recoils away from the wrench. (b) the distance covered by the astronaut in 30 minutes. (Book's printed answers: -1.5 x 10^-2 m s^-1, 27 m)

(a) Step 1: wrench mass = 300 g = 0.300 kg. Step 2: total momentum before is zero, so 0 = 0.300 x 3.5 + 70 x v. Step 3: 70 v = -1.05, so v = -0.015 m s^-1 = -1.5 x 10^-2 m s^-1, the minus sign showing recoil away from the wrench. (b) Step 1: time = 30 minutes = 1800 s. Step 2: distance = speed x time = 0.015 x 1800 = 27 m. Both match the book's printed answers.

E. Numerical Problems

3.6 A cyclist weighing 55 kg rides a bicycle of mass 5 kg. He starts from rest and applies a force of 90 N for 8 seconds. Then he continues at a constant speed for another 8 seconds. Calculate the total distance travelled by the cyclist. (Book's printed answer: 144 m)

Step 1: total mass = 55 + 5 = 60 kg. Step 2: acceleration a = F/m = 90/60 = 1.5 m s^-2. Step 3: distance in the first 8 s: S1 = (1/2) a t^2 = (1/2) x 1.5 x 64 = 48 m. Step 4: speed reached: v = a t = 1.5 x 8 = 12 m s^-1. Step 5: distance at constant speed for 8 s: S2 = 12 x 8 = 96 m. Step 6: total distance = 48 + 96 = 144 m. This matches the book's printed answer.

E. Numerical Problems

3.7 A ball of mass 0.4 kg is dropped on the floor from a height of 1.8 m. The ball rebounds straight upward to a height of 0.8 m. What is the magnitude and

direction of the impulse applied to the ball by the floor? (Book's printed answer: 4 N s, upward)

Step 1: speed just before hitting the floor: v = sqrt(2 g h) = sqrt(2 x 10 x 1.8) = sqrt(36) = 6 m s^-1 downwards. Step 2: speed just after rebound: v' = sqrt(2 x 10 x 0.8) = sqrt(16) = 4 m s^-1 upwards. Step 3: taking upward as positive, change in velocity = 4 - (-6) = 10 m s^-1. Step 4: impulse = m x change in velocity = 0.4 x 10 = 4 N s, directed upward. This matches the book's printed answer.

E. Numerical Problems

3.8 Two balls of masses 0.2 kg and 0.4 kg are moving towards each other with velocities 20 m s^-1 and 5 m s^-1 respectively. After collision, the velocity of 0.2 kg ball becomes 6 m s^-1. What will be the velocity of 0.4 kg ball? (Book's printed answer: 2 m s^-1)

Step 1: take the 0.2 kg ball's direction as positive. Initial total momentum = 0.2 x 20 + 0.4 x (-5) = 4 - 2 = 2 kg m s^-1. Step 2: final total momentum = 0.2 x 6 + 0.4 x v = 1.2 + 0.4 v. Step 3: by conservation of momentum, 1.2 + 0.4 v = 2, so 0.4 v = 0.8 and v = 2 m s^-1. The positive sign means the 0.4 kg ball has reversed direction and now moves with the 0.2 kg ball at 2 m s^-1. This matches the book's printed answer.

Chapter 4: Turning Effects of Force

A. Multiple Choice Questions

4.1 A particle is simultaneously acted upon by two forces of 4 and 3 newtons. The net force on the particle is:

(a) 1 N

(b) between 1 N and 7 N ✓

(c) 5 N

(d) 7 N

Correct answer: between 1 N and 7 N

Working for Question 4.1

Working: range of the net force

The net force depends on the angle between the two forces. If they act in opposite directions it is 4 - 3 = 1 N; if they act in the same direction it is 4 + 3 = 7 N; at any other angle it lies between these values. So the net force is between 1 N and 7 N, option (b). Only if the forces were perpendicular would it be exactly 5 N, which the question does not state.

A. Multiple Choice Questions

4.2 If F1 and F2 are the forces acting on a body and T is the torque produced in it, the body will be completely in equilibrium, when:

(a) sum F = 0 and sum T = 0 ✓

(b) sum F = 0 and sum T not = 0

(c) sum F not = 0 and sum T = 0

(d) sum F not = 0 and sum T not = 0

Correct answer: sum F = 0 and sum T = 0

A. Multiple Choice Questions

4.3 A Shopkeeper sells his articles by a balance having unequal arms of the pans. If he puts the weights in the pan having shorter arm, then the customer:

(a) loses ✓

(b) gains

(c) neither loses nor gains

(d) not certain

Correct answer: loses

Working for Question 4.3

Working: why does the customer lose?

For balance, weight x short arm = goods x long arm. Since the short arm is smaller than the long arm, the goods weighed are less than the weights shown: goods = weights x (short arm / long arm), which is less than the weights. The customer therefore receives less goods than he pays for, so he loses, option (a).

A. Multiple Choice Questions

4.4 A man walks on a tight rope. He balances himself by holding a bamboo stick horizontally. It is an application of:

(a) law of conservation of momentum

(b) Newton's second law of motion

(c) principle of moments ✓

(d) Newton's third law of motion

Correct answer: principle of moments

A. Multiple Choice Questions

4.5 The centre of mass of a body:

(a) lies always inside the body

(b) lies always outside the body

(c) lies always on the surface of the body

(d) may lie within, outside or on the surface ✓

Correct answer: may lie within, outside or on the surface

A. Multiple Choice Questions

4.6 A cylinder resting on its circular base is in:

(a) stable equilibrium ✓

(b) unstable equilibrium

(c) neutral equilibrium

(d) none of these

Correct answer: stable equilibrium

B. Short Answer Questions

4.1 What are rectangular components of a vector and their values?

Rectangular components are the two perpendicular parts into which a vector is resolved, usually along the x-axis and the y-axis. For a vector F making an angle theta with the x-axis, the x-component is Fx = F cos(theta) and the y-component is Fy = F sin(theta). Together they completely replace the original vector: F is the resultant of Fx and Fy.

B. Short Answer Questions

4.2 What is the line of action of a force?

The line of action of a force is the straight line drawn through the point where the force acts, in the direction of the force, extended in both directions. The turning effect of the force about a point depends on the perpendicular distance from that point to this line.

B. Short Answer Questions

4.3 Define moment of a force. Prove that T = r F sin(theta), where theta is angle between r and F.

The moment, or torque, of a force is the turning effect of the force about a point, equal to the force multiplied by the perpendicular distance from the point to the line of action of the force. Proof: let r be the distance from the pivot to the point of application, and let the force F make angle theta with r. The component of F perpendicular to r is F sin(theta). The perpendicular distance from the pivot to the line of action is r sin(theta). Hence torque T = F x r sin(theta) = r F sin(theta). When the force is perpendicular to r, theta = 90 degrees and T = r F.

B. Short Answer Questions

4.4 With the help of a diagram, show that the resultant force is zero but the resultant torque is not zero.

This is a couple: two equal and opposite parallel forces acting on a body. Diagram to draw: a horizontal rod with an upward force F at one end and a downward force F at the other end, separated by distance d. The forces cancel, so the resultant force is zero and the rod does not move bodily. But both forces turn the rod in the same rotational direction, so their moments add: total torque = F x d, which is not zero, and the rod rotates.

B. Short Answer Questions

4.5 Identify the state of equilibrium in each case in the figure given below. (a) cone on its base (b) ball on a dome (c) cylinder lying on its side

(a) The cone standing on its flat base is in stable equilibrium: if tilted slightly, its centre of gravity rises and gravity pulls it back to its original position. (b) The ball balanced on top of a dome is in unstable equilibrium: the slightest displacement lowers its centre of gravity and it rolls away and does not return. (c) The cylinder lying on its side is in neutral equilibrium: rolling it to a new

position leaves its centre of gravity at the same height, so it stays wherever it is put.

B. Short Answer Questions

4.6 Give an example of the body which is moving yet in equilibrium.

A car moving along a straight road at a constant velocity is in equilibrium even though it is moving. Its forces balance (driving force equals friction and air resistance) and its moments balance, so the net force and net torque are both zero. This is dynamic equilibrium: equilibrium does not require rest, only zero net force and zero net torque.

B. Short Answer Questions

4.7 What are two basic principles of stability in physics which are applied in designing balancing toys and racing cars?

The two principles are: (1) keep the centre of gravity as low as possible, and (2) make the base of support as wide as possible. Racing cars are built low and wide for exactly this reason, and balancing toys have a heavy base so that their centre of gravity stays low and within the base.

B. Short Answer Questions

4.8 How can you prove that the centripetal force always acts perpendicular to velocity?

In uniform circular motion the speed of the body stays constant, so the force cannot have any component along the velocity, because such a component would change the speed by doing work. The force can only change the direction of the velocity, and a change of direction with no change of speed is produced by a force perpendicular to the velocity. Hence the centripetal force acts perpendicular to the velocity, towards the centre.

C. Constructed Response Questions

4.1 A car travels at the same speed around two curves with different radii. For which radius the car experiences more centripetal force? Prove your answer.

The car experiences more centripetal force on the curve with the smaller radius. Proof: centripetal force F = m v^2 / r. Mass m and speed v are the same for both curves, so F is inversely proportional to the radius r. Halving the radius doubles the force. Therefore the tighter curve, with the smaller radius, needs the greater centripetal force to keep the car on the road.

C. Constructed Response Questions

4.2 A ripe mango does not normally fall from the tree. But when the branch of the tree is shaken, the mango falls down easily. Can you tell the reason?

The mango is held to the branch by its stem, which is strong enough to support its weight at rest. When the branch is shaken, the branch moves suddenly but the mango, by inertia, tends to stay where it was. The stem is jerked and weakened, and the repeated jerks break it, so the mango falls. Shaking therefore uses the mango's inertia to snap the stem that otherwise holds it.

C. Constructed Response Questions

4.3 Discuss the concepts of stability and centre of gravity in relation to objects toppling over. Provide an example where an object's centre of gravity affects its stability, and explain how altering its base of support can influence stability.

An object topples when the vertical line through its centre of gravity falls outside its base of support. A tall, narrow object has a high centre of gravity, so a small tilt moves that line outside the base and it topples easily, for example a tall bookcase. A low, wide object is stable because its centre of gravity stays above the base even when tilted. Widening the base of support, for example by fitting wider feet or outriggers, keeps the vertical line through the centre of gravity

inside the base for larger tilts, so the object becomes more stable and harder to topple.

C. Constructed Response Questions

4.4 Why an accelerated body cannot be considered in equilibrium?

Equilibrium requires the net force on a body to be zero, by the first condition of equilibrium. But an accelerated body has a net force acting on it, since F = m a, and a is not zero. A body with a net force cannot satisfy the equilibrium conditions, so it cannot be considered in equilibrium, even if it happens to be momentarily at rest.

C. Constructed Response Questions

4.5 Two boxes of the same weight but different heights are lying on the floor of a truck. If the truck makes a sudden stop, which box is more likely to tumble over? Why?

The taller box is more likely to tumble over. When the truck stops suddenly, the bottom of each box stops with the truck but the top of each box tends to keep moving forward by inertia. The taller box has a higher centre of gravity, so the forward toppling moment about its front bottom edge is greater and the vertical line through its centre of gravity leaves its base more easily. The shorter box, with its lower centre of gravity, is more stable and more likely to stay standing or merely slide.

D. Comprehensive Questions

4.1 Explain the principle of moments with an example.

The principle of moments states that if a body is in equilibrium, the sum of the clockwise moments about any point equals the sum of the anticlockwise moments about that point. Example: a see-saw balances when a heavier child

sits near the pivot and a lighter child sits farther away, because weight x distance is the same on both sides. If the clockwise moment exceeds the anticlockwise moment, the body rotates in the clockwise direction.

D. Comprehensive Questions

4.2 Describe how could you determine the centre of gravity of an irregular shaped lamina experimentally.

Hang the lamina freely from a pin through a hole near its edge and let it settle. Hang a plumb line from the same pin and mark the vertical line on the lamina. Repeat by hanging the lamina from a second, different hole and marking the new vertical line. The point where the two lines cross is the centre of gravity, because a freely suspended body always hangs with its centre of gravity vertically below the point of suspension. A third suspension can be used to check the point.

D. Comprehensive Questions

4.3 State and explain two conditions of equilibrium.

First condition: the vector sum of all forces on the body must be zero, so the body has no linear acceleration and stays at rest or moves uniformly. Second condition: the sum of all torques (moments) about any point must be zero, so the body has no angular acceleration and does not start to rotate. Both conditions must hold together: a couple has zero net force but a non-zero torque, so it fails the second condition and is not in equilibrium.

D. Comprehensive Questions

4.4 How the stability of an object can be improved? Give a few examples to support your answer.

Stability is improved by lowering the centre of gravity and by widening the base

of support. Examples: racing cars are built low and wide; a table lamp or a laboratory stand has a heavy, broad base; a tightrope walker carries a long pole to widen the effective base and lower the centre of gravity; and a ship carries heavy cargo low in its hold so that it rights itself after rolling.

E. Numerical Problems

4.1 A force of 200 N is acting on a cart at an angle of 30 degrees with the horizontal direction. Find the x and y-components of the force. (Book's printed answers: 173.2 N, 100 N)

Step 1: x-component Fx = F cos(30 degrees) = 200 x 0.866 = 173.2 N. Step 2: y-component Fy = F sin(30 degrees) = 200 x 0.5 = 100 N. Both match the book's printed answers.

E. Numerical Problems

4.2 Two weights are hanging from a metre rule at the positions as shown in the given figure. If the metre rule is balanced at its centre of gravity (C.G), find the unknown weight w. (Book's printed answer: 3 N)

Step 1: from the figure, the unknown weight w hangs 40 cm to the left of the C.G, and the 4 N weight hangs 30 cm to the right. Step 2: by the principle of moments, anticlockwise moment = clockwise moment, so w x 40 = 4 x 30. Step 3: w = 120/40 = 3 N. This matches the book's printed answer.

E. Numerical Problems

4.3 A see-saw is balanced with two children sitting near either end. Child A weighs 30 kg and sits 2 metres away from the pivot, while child B weighs 40 kg and sits 1.5 metres from the pivot. Calculate the total moment on each side and determine if the sea-saw is in equilibrium. (Book's printed answer: 60 N)

Step 1: weight of A = 30 x 10 = 300 N; moment of A = 300 x 2 = 600 N m. Step

2: weight of B = 40 x 10 = 400 N; moment of B = 400 x 1.5 = 600 N m. Step 3: the moments are equal (600 N m each side), so the clockwise and anticlockwise moments balance and the see-saw is in equilibrium. Note on the book's printed answer: the book prints (60 N), which is the mass x distance product on each side (30 x 2 = 60 and 40 x 1.5 = 60); with weights included the moments are 600 N m each side, and the conclusion, equilibrium, is the same.

E. Numerical Problems

4.4 A crowbar is used to lift a box as shown in the given figure. If the downward force of 250 N is applied at the end of the bar, how much weight does the other end bear? The crowbar itself has negligible weight. (Book's printed answer: 1500 N)

Step 1: from the figure, the effort arm is 30 cm and the load arm is 5 cm from the fulcrum. Step 2: by the principle of moments, load x load arm = effort x effort arm, so W x 5 = 250 x 30. Step 3: W = 7500/5 = 1500 N. This matches the book's printed answer.

E. Numerical Problems

4.5 A 30 cm long spanner is used to open the nut of a car. If the torque required for it is 150 N m, how much force F should be applied on the spanner as shown in the figure given below. (Book's printed answer: 500 N)

Step 1: length of spanner = 30 cm = 0.30 m. Step 2: torque = force x perpendicular distance, so 150 = F x 0.30. Step 3: F = 150/0.30 = 500 N. This matches the book's printed answer.

E. Numerical Problems

4.6 A signboard is suspended by means of two steel wires as shown in the given figure. If the weight of the board is 200 N, what is the tension in the strings?

(Book's printed answers: 100 N, 100 N)

Step 1: the board is symmetric, hung 2 m on each side of its centre, so the two wires share the weight equally. Step 2: total upward force = total downward weight, so T1 + T2 = 200 N. Step 3: by symmetry T1 = T2, so each tension = 200/2 = 100 N. Both match the book's printed answers.

E. Numerical Problems

4.7 One girl of 30 kg mass sits 1.6 m from the axis of a see-saw. Another girl of mass 40 kg wants to sit on the other side, so that the see-saw may remain in equilibrium. How far away from the axis, the other girl may sit? (Book's printed answer: 1.2 m)

Step 1: by the principle of moments, 30 x 1.6 = 40 x d. Step 2: 48 = 40 d, so d = 48/40 = 1.2 m. This matches the book's printed answer. (Masses are used on both sides, so the factor g cancels.)

E. Numerical Problems

4.8 Find the tension in each string as shown in the given figure, if the block weighs 150 N. (Book's printed answers: 86.6 N, 173.2 N)

Step 1: read the figure as a block hung from a junction of two strings, one making 60 degrees with the horizontal and the other horizontal. Step 2: vertical equilibrium at the junction: the vertical component of the slanted string balances the weight, so T x sin(60 degrees) = 150 N. Step 3: T = 150/0.866 = 173.2 N for the slanted string. Step 4: horizontal equilibrium: the horizontal string balances the horizontal component, so T_horizontal = 173.2 x cos(60 degrees) = 173.2 x 0.5 = 86.6 N. Both match the book's printed answers.

Chapter 5: Work, Energy and Power

A. Multiple Choice Questions

5.1 Work done is maximum when the angle between the force F and the displacement d is:

(a) 0 degrees ✓

(b) 30 degrees

(c) 60 degrees

(d) 90 degrees

Correct answer: 0 degrees

A. Multiple Choice Questions

5.2 A joule can also be written as:

(a) kg m s^-2

(b) kg m s^-1

(c) kg m^2 s^-3

(d) kg m^2 s^-2 ✓

Correct answer: kg m^2 s^-2

A. Multiple Choice Questions

5.3 The power of a water pump is 2 kW. The amount of water it can raise in one minute to a height of 5 metres is:

(a) 1000 litres

(b) 1200 litres

(c) 2000 litres

(d) 2400 litres ✓

Correct answer: 2400 litres

Working for Question 5.3

Working: water raised by a 2 kW pump in one minute

Step 1: energy supplied in one minute = power x time = 2000 x 60 = 120,000 J.

Step 2: energy needed to raise mass m through 5 m is m g h = m x 10 x 5 = 50 m. Step 3: 50 m = 120,000, so m = 2400 kg. Step 4: 1 kg of water is 1 litre, so the pump raises 2400 litres, option (d).

A. Multiple Choice Questions

5.4 A bullet of mass 0.05 kg has a speed of 300 m s^-1. Its kinetic energy will be:

(a) 2250 J ✓

(b) 4500 J

(c) 1500 J

(d) 1125 J

Correct answer: 2250 J

Working for Question 5.4

Working: kinetic energy of the bullet

Kinetic energy = (1/2) m v^2 = (1/2) x 0.05 x (300)^2 = (1/2) x 0.05 x 90,000 = 2250 J. So option (a) is correct.

A. Multiple Choice Questions

5.5 If a car doubles its speed, its kinetic energy will be:

(a) the same

(b) doubled

(c) increased to three times

(d) increased to four times ✓

Correct answer: increased to four times

A. Multiple Choice Questions

5.6 The magnitude of momentum of an object is doubled, the kinetic energy of the object will:

(a) double

(b) increase to four times ✓

(c) reduce to one-half

(d) remain the same

Correct answer: increase to four times

Working for Question 5.6

Working: why does doubling momentum quadruple kinetic energy?

Kinetic energy can be written as K.E. = p^2 / (2 m), where p is momentum. If p doubles while m stays the same, p^2 becomes four times as large, so the kinetic energy increases to four times, option (b).

A. Multiple Choice Questions

5.7 Which of the following is not renewable energy source?

(a) Hydroelectric energy

(b) Fossil fuels ✓

(c) Wind energy

(d) Solar energy

Correct answer: Fossil fuels

B. Short Answer Questions

5.1 A slow-moving car may have more kinetic energy than a fast-moving motorcycle. How is this possible?

Kinetic energy depends on both mass and speed: K.E. = (1/2) m v^2. A car is far heavier than a motorcycle, so even at a lower speed its large mass can give it

more kinetic energy than a light motorcycle moving faster. The mass factor outweighs the speed factor in this comparison.

B. Short Answer Questions

5.2 A force F1 does 5 J of work in 10 s. Another force F2 does 3 J of work in 5 s. Which force delivers greater power?

Power = work / time. Power of F1 = 5/10 = 0.5 W. Power of F2 = 3/5 = 0.6 W. F2 delivers the greater power, even though it does less total work, because it does its work in a much shorter time.

B. Short Answer Questions

5.3 A woman runs up a flight of stairs. The gain in her gravitational potential energy is 4500 J. If she runs up the same stairs with twice the speed, what will be her gain in potential energy?

Still 4500 J. Gravitational potential energy depends only on weight and on the vertical height gained, P.E. = m g h. The height of the stairs does not change, so the gain in potential energy is the same; running faster only means she gains that energy in a shorter time, that is at a greater power.

B. Short Answer Questions

5.4 What is power? Define the unit used for it.

Power is the rate of doing work, that is work done per unit time: P = W / t. Its SI unit is the watt (W). One watt is the power when one joule of work is done in one second: 1 W = 1 J s^-1. Larger units are the kilowatt (1 kW = 1000 W) and the horsepower.

B. Short Answer Questions

5.5 Differentiate between renewable and non-renewable energy sources.

Renewable energy sources are naturally replenished as fast as they are used, for example solar energy, wind energy, hydroelectric energy and geothermal energy; they do not run out and generally cause little pollution. Non-renewable energy sources exist in a fixed stock that is used up far faster than nature can replace it, for example coal, oil, natural gas and uranium; once used, they are gone and their use usually produces pollution.

C. Constructed Response Questions

5.1 Can the kinetic energy of a body ever be negative?

No. Kinetic energy is (1/2) m v^2. Mass m is always positive, and v^2 is always positive or zero whatever the direction of motion, because squaring removes the sign. The product is therefore never negative; the smallest possible kinetic energy is zero, for a body at rest.

C. Constructed Response Questions

5.2 Which one has the greater kinetic energy; an object travelling with a velocity v or an object twice as heavy travelling with a velocity of 1/2 v?

The first object has the greater kinetic energy. First object: K.E. = (1/2) m v^2. Second object: K.E. = (1/2) x (2 m) x (v/2)^2 = (1/2) x 2 m x v^2/4 = (1/4) m v^2. Comparing, (1/2) m v^2 is double (1/4) m v^2, so the lighter, faster object has twice the kinetic energy of the heavier, slower one.

C. Constructed Response Questions

5.3 While playing cricket on a street, the ball smashes a window pane. Describe the energy changes in this event.

Before impact the ball has kinetic energy. On hitting the pane, that kinetic energy is transformed: part of it breaks the chemical bonds of the glass, part becomes sound energy, part becomes heat energy in the glass and the ball, part

becomes kinetic energy of the flying glass pieces, and the ball keeps a little kinetic energy as it rebounds or drops. The total energy stays constant; it is only converted from one form to another.

C. Constructed Response Questions

5.4 A man rowing boat upstream is at rest with respect to the shore. Is he doing work?

In the physics sense, no work is done on the boat, because work = force x displacement, and the boat's displacement relative to the shore is zero: it stays in one place. The man certainly gets tired, because his muscles expend chemical energy internally, but since the boat does not move, no mechanical work is done on it.

C. Constructed Response Questions

5.5 A cyclist goes downhill from the top of a steep hill without pedalling and takes it to the top of the next hill. (i) Draw a diagram of what happened. (ii) Analyse this event in terms of potential and kinetic energy. Label your diagram using these terms.

(i) Diagram to draw: two hills side by side, the first taller than the second, with a valley between them; a cyclist path runs down the first hill, across the valley and part-way up the second hill. (ii) Analysis: at the top of the first hill the cyclist has maximum gravitational potential energy and little kinetic energy. Rolling downhill, potential energy converts into kinetic energy, which is maximum at the bottom of the valley. Climbing the second hill, kinetic energy converts back into potential energy, so the cyclist rises until the remaining kinetic energy is used up. Label the first hilltop 'maximum P.E., minimum K.E.', the valley 'maximum K.E., minimum P.E.', and the stopping point on the second hill 'P.E. gained, K.E. zero'. Because of friction, the second hill reached is lower than the first.

C. Constructed Response Questions

5.6 Is timber or wood renewable source of heat energy? Comment.

Yes, but only conditionally. Timber is renewable if trees are replanted and allowed to regrow at least as fast as they are cut, because the wood is then replaced by nature on a human timescale. If forests are cut faster than they regrow, the supply runs down just like a non-renewable source, and the loss of forest causes other damage. Timber is therefore a renewable source only under sustainable forestry.

D. Comprehensive Questions

5.1 What is meant by kinetic energy? State its unit. Describe how it is determined.

Kinetic energy is the energy a body has because of its motion. Its SI unit is the joule (J). It is determined by the work done to accelerate the body from rest to its velocity: a constant force F acting over a distance S does work F x S, and using F = m a and v^2 = 2 a S, this work equals (1/2) m v^2. So kinetic energy = (1/2) m v^2, where m is the mass and v is the velocity.

D. Comprehensive Questions

5.2 State the law of conservation of energy. Explain it with the help of an example of a body falling from certain height in terms of its potential energy and kinetic energy.

The law of conservation of energy states that energy can neither be created nor destroyed; it can only be changed from one form to another, and the total energy of an isolated system stays constant. Example: a body held at a height has gravitational potential energy m g h and zero kinetic energy. As it falls, its height decreases so its potential energy decreases, while its speed increases so its kinetic energy increases by exactly the same amount. Just before hitting the

ground its potential energy is zero and its kinetic energy is m g h. At every point, potential energy + kinetic energy = m g h, a constant.

D. Comprehensive Questions

5.3 Explain what is meant by efficiency of a machine. How is it calculated? Why there is a limit for the efficiency of a machine?

Efficiency is the ratio of the useful work or energy a machine gives out to the total work or energy put into it, expressed as a percentage: percentage efficiency = (useful output energy / total input energy) x 100. There is a limit because no machine is perfect: part of the input energy is always wasted, mainly as heat due to friction between moving parts, and as sound. Since the useful output is always less than the input, efficiency is always less than 100 percent.

E. Numerical Problems

5.1 A force of 20 N acting at an angle of 60 degrees to the horizontal is used to pull a box through a distance of 3 m across a floor. How much work is done? (Book's printed answer: 30 J)

Step 1: work W = F d cos(theta). Step 2: W = 20 x 3 x cos(60 degrees) = 60 x 0.5 = 30 J. This matches the book's printed answer.

E. Numerical Problems

5.2 An engine raises 100 kg of water through a height of 80 m in 25 s. What is the power of the engine? (Book's printed answer: 3200 W)

Step 1: work done = gain in potential energy = m g h = 100 x 10 x 80 = 80,000 J. Step 2: power P = work / time = 80,000 / 25 = 3200 W. This matches the book's printed answer.

E. Numerical Problems

5.3 A body of mass 20 kg is at rest. A 40 N force acts on it for 5 seconds. What is the kinetic energy of the body at the end of this time? (Book's printed answer: 1000 J)

Step 1: acceleration a = F/m = 40/20 = 2 m s^-2. Step 2: velocity after 5 s: v = a t = 2 x 5 = 10 m s^-1. Step 3: kinetic energy = (1/2) m v^2 = (1/2) x 20 x 100 = 1000 J. This matches the book's printed answer.

E. Numerical Problems

5.4 A 0.14 kg ball is thrown vertically upward with an initial velocity of 35 m s^-1. Find the maximum height reached by the ball. (Book's printed answer: 61.25 m)

Step 1: at maximum height, final velocity vf = 0. Step 2: use vf^2 = vi^2 - 2 g h, so 0 = (35)^2 - 2 x 10 x h. Step 3: 20 h = 1225, so h = 1225/20 = 61.25 m. This matches the book's printed answer.

E. Numerical Problems

5.5 A girl is swinging on a swing. At the lowest point of her swing, she is 1.2 m from the ground, and at the highest point she is 2.0 m from the ground. What is her maximum velocity and where? (Book's printed answer: 4 m s^-1, at the lowest position)

Step 1: the height difference between the highest and lowest points is 2.0 - 1.2 = 0.8 m. Step 2: by conservation of energy, the potential energy lost in falling 0.8 m becomes kinetic energy: (1/2) m v^2 = m g x 0.8. Step 3: v^2 = 2 x 10 x 0.8 = 16, so v = 4 m s^-1. Step 4: this maximum velocity occurs at the lowest position, where the kinetic energy is greatest. This matches the book's printed answer.

E. Numerical Problems

5.6 A person pushes a lawn mower with a force of 50 N making an angle of 45 degrees with the horizontal. If the mower is moved through a distance of 20 m, how much work is done? (Book's printed answer: 707 J)

Step 1: work W = F d cos(theta) = 50 x 20 x cos(45 degrees). Step 2: cos(45 degrees) = 0.707. Step 3: W = 1000 x 0.707 = 707 J. This matches the book's printed answer.

E. Numerical Problems

5.7 A box of mass 10 kg is pushed up along a ramp 15 m long with a force of 80 N. If the box rises up a height of 5 m, what is the efficiency of the system? (Book's printed answer: 41.7 percent)

Step 1: input work = force x distance along ramp = 80 x 15 = 1200 J. Step 2: useful output work = gain in potential energy = m g h = 10 x 10 x 5 = 500 J. Step 3: efficiency = (500/1200) x 100 = 41.7 percent. This matches the book's printed answer.

E. Numerical Problems

5.8 A force of 600 N acts on a box to push it 5 m in 15 s. Calculate the power. (Book's printed answer: 200 W)

Step 1: work done = force x distance = 600 x 5 = 3000 J. Step 2: power = work / time = 3000/15 = 200 W. This matches the book's printed answer.

Chapter 6: Mechanical Properties of Matter

A. Multiple Choice Questions

6.1 A wire is stretched by a weight w. If the diameter of the wire is reduced to half of its previous value, the extension will become:

(a) one half

(b) double

(c) one fourth

(d) four times ✓

Correct answer: four times

Working for Question 6.1

Working: extension when diameter is halved

Extension is inversely proportional to the cross-sectional area, and area is proportional to the square of the diameter. Halving the diameter reduces the area to one-fourth, so the extension becomes four times as large, option (d).

A. Multiple Choice Questions

6.2 Four wires of the same material are stretched by the same load. Their dimensions are given below. Which of them will elongate most?

(a) Length 1 m, diameter 1 mm

(b) Length 2 m, diameter 2 mm

(c) Length 3 m, diameter 3 mm

(d) Length 4 m, diameter 0.5 mm ✓

Correct answer: Length 4 m, diameter 0.5 mm

Working for Question 6.2

Working: which wire elongates most?

Extension is proportional to length divided by area, and area is proportional to

diameter squared, so extension is proportional to length / diameter^2. The values are: (a) 1/1 = 1; (b) 2/4 = 0.5; (c) 3/9 = 0.33; (d) 4/0.25 = 16. Wire (d) has by far the largest value, so it elongates most, option (d).

A. Multiple Choice Questions

6.3 Two metal plates of area 2 and 3 square metres are placed in a liquid at the same depth. The ratio of pressures on the two plates is:

(a) 1:1

(b) sqrt(2) : sqrt(3)

(c) 2:3

(d) 4:9

Correct answer: 1:1

A. Multiple Choice Questions

6.4 The pressure at any point in a liquid is proportional to:

(a) density of the liquid

(b) depth of the point below the surface of the liquid

(c) acceleration due to gravity

(d) all of the above

Correct answer: all of the above

A. Multiple Choice Questions

6.5 Pressure applied to an enclosed fluid is:

(a) increased and applied to every part of the fluid

(b) diminished and transmitted to the walls of container

(c) increased in proportional to the mass of fluid and then transmitted to each part of the fluid

(d) transmitted unchanged to every portion of the fluid and walls of containing

vessel

Correct answer: transmitted unchanged to every portion of the fluid and walls of containing vessel

A. Multiple Choice Questions

6.6 What is the force exerted by the atmosphere on a rectangular block surface of length 50 cm and breadth 40 cm? The atmospheric pressure is 100 kPa.

(a) 20 kN

(b) 100 kN

(c) 200 kN

(d) 500 kN

Correct answer: 20 kN

Working for Question 6.6

Working: force of the atmosphere on the block

Step 1: area = 0.50 m x 0.40 m = 0.20 m^2. Step 2: pressure = 100 kPa = 100,000 Pa. Step 3: force = pressure x area = 100,000 x 0.20 = 20,000 N = 20 kN. So option (a) is correct.

B. Short Answer Questions

6.1 Why heavy animals like an elephant have a large area of the foot?

Pressure equals force divided by area. An elephant is very heavy, so the force of its weight is huge. A large foot area spreads that weight over a larger area, keeping the pressure on the ground low, so the elephant does not sink deeply into soft ground and its feet are not injured.

B. Short Answer Questions

6.2 Why animals like deer who run fast have a small area of the foot?

A small foot area gives a high pressure for the same weight, so the hoof presses firmly into the ground and grips it instead of slipping. That grip is what lets a deer push off powerfully and run fast. A small, hard hoof also lifts and moves quickly, which suits fast running.

B. Short Answer Questions

6.3 Why is it painful to walk bare footed on pebbles?

A pebble touches the foot over a very tiny area, so the whole weight of the body presses through that small area and the pressure becomes very large. That large pressure on a small spot of skin is what causes the pain.

B. Short Answer Questions

6.4 What is Hooke's Law? Does an object remain elastic beyond elastic limit? Give reason.

Hooke's law states that within the elastic limit, the extension of a spring or wire is directly proportional to the stretching force applied to it. No, beyond the elastic limit the object does not remain elastic: it suffers a permanent deformation and does not return to its original length when the force is removed, because its internal structure has been permanently changed.

B. Short Answer Questions

6.5 Distinguish between force and pressure.

Force is a push or pull acting on a body, measured in newtons (N); it can change motion or shape. Pressure is force per unit area, measured in pascals (Pa = N m^-2); it describes how concentrated a force is. The same force produces a large pressure on a small area and a small pressure on a large area, which is why a sharp knife cuts while a blunt one does not.

B. Short Answer Questions

6.6 What is the relationship between liquid pressure and the depth of the liquid?

Liquid pressure increases in direct proportion to depth: P = rho g h, where rho is the density of the liquid, g is the acceleration due to gravity and h is the depth below the surface. Doubling the depth doubles the pressure at that point.

B. Short Answer Questions

6.7 What is the basic principle to measure the atmospheric pressure by a simple mercury barometer?

The atmosphere presses on the mercury in the open reservoir, and that pressure supports the column of mercury in the evacuated tube above it. The height of the column therefore balances the atmospheric pressure: a higher atmospheric pressure supports a taller column. Measuring the column height gives the atmospheric pressure directly.

B. Short Answer Questions

6.8 State the basic principle used in the hydraulic brake system of the automobiles.

Pascal's law: pressure applied to an enclosed liquid is transmitted undiminished to every part of the liquid and to the walls of the container. Pressing the brake pedal creates pressure in the brake fluid, and that same pressure acts on the larger pistons at the wheels, producing the large force that presses the brake pads against the discs.

C. Constructed Response Questions

6.1 A spring having spring constant k hangs vertically from a fixed point. A load of weight L, when hung from the spring, causes an extension x, the elastic limit of the spring is not exceeded. Some identical springs, each of spring constant k,

are arranged as shown below: (1) two springs in series, one below the other (2) two springs in parallel, side by side. For each arrangement, complete the table by determining: (i) the total extension in terms of x. (ii) the spring constant in terms of k.

Single spring: extension x, constant k, since L = k x. (1) Series arrangement: each spring carries the full load L, so each extends by x, and the total extension is x + x = 2x. The effective constant is L / (2x) = k/2. (2) Parallel arrangement: the load is shared equally, so each spring carries L/2 and extends by x/2; the total extension is therefore x/2. The effective constant is L / (x/2) = 2k. In short: series gives extension 2x and constant k/2; parallel gives extension x/2 and constant 2k.

C. Constructed Response Questions

6.2 Springs are made of steel instead of iron. Why?

Steel is more elastic than iron: it has a higher elastic limit and a larger spring constant, so a steel spring stretches under load and returns fully to its original shape, again and again, without taking a permanent set. Iron is softer and more easily deformed permanently, so an iron spring would soon lose its shape and stop working as a spring.

C. Constructed Response Questions

6.3 How does water pressure one metre below the surface of a swimming pool compare to water pressure one metre below the surface of a very large and deep lake?

They are exactly equal. Liquid pressure at a point depends only on the density of the liquid, the depth of the point and g: P = rho g h. The size of the pool or lake, and the total volume of water in it, do not appear in the formula, so at the same depth of 1 m in the same liquid the pressure is the same in both.

C. Constructed Response Questions

6.4 What will happen to the pressure in all parts of a confined liquid if pressure is increased in one part? Give an example from your daily life where such principle is applied.

By Pascal's law, the increase in pressure is transmitted undiminished to all parts of the confined liquid and to the walls of the container. Everyday examples are the hydraulic brake of a car, where pressing the pedal raises the pressure everywhere in the brake fluid, and a hydraulic car lift, where a small force on a small piston lifts a heavy car on a large piston.

C. Constructed Response Questions

6.5 If some air remains trapped within the top of the mercury column of the barometer which is supposed to be vacuum, how would it affect the height of the mercury column?

The trapped air exerts its own downward pressure on top of the mercury column, partly balancing the atmospheric pressure from below. The mercury column therefore does not need to rise as high to balance the atmosphere, so the column is shorter than it should be and the barometer reads a pressure lower than the true atmospheric pressure.

C. Constructed Response Questions

6.6 The end of glass tube used in a simple barometer is not properly sealed, some leak is present. What will be its effect?

Air leaks into the supposed vacuum above the mercury, so the space above the column is no longer empty. The leaked air presses down on the mercury and the column falls; if the leak is large the mercury level drops to the reservoir level and the barometer stops working altogether, because it can no longer balance the atmospheric pressure.

C. Constructed Response Questions

6.7 Comment on the statement. 'Density is a property of a material not the property of an object made of that material.'

The statement is correct. Density is mass divided by volume, and for a given material this ratio is the same whatever the size or shape of the object: a small iron nail and a large iron block both have the density of iron, about 7.8 x 10^3 kg m^-3, even though their masses and volumes differ greatly. Mass and volume are properties of the object; density is a property of the material itself.

C. Constructed Response Questions

6.8 How the load of a large structure is estimated by an engineer?

The engineer adds up the weights of all the parts: for each part, weight = volume x density x g, using the known densities of steel, concrete, brick and other materials. To this dead load are added the live loads the structure must carry, such as people, furniture, vehicles and wind pressure, plus a safety factor. The total gives the load the foundations and supports must be designed to bear, and it also fixes the pressure the structure exerts on the ground.

D. Comprehensive Questions

6.1 What is Hooke's law? Give three applications of this law.

Hooke's law states that within the elastic limit, the extension of an elastic body is directly proportional to the force stretching it: F = k x, where k is the spring constant. Three applications: (1) spring balances, which measure weight or force from the extension of a spring; (2) vehicle suspension springs, which absorb shocks by stretching and compressing; (3) bathroom and kitchen weighing scales, which convert the compression of a spring into a weight reading.

D. Comprehensive Questions

6.2 Describe the working and applications of a simple mercury barometer and a manometer.

Simple mercury barometer: a glass tube filled with mercury is inverted into a mercury reservoir. The atmosphere presses on the reservoir and supports a mercury column about 76 cm high; the height measures the atmospheric pressure. It is used to measure atmospheric pressure and to forecast weather, since falling pressure often means rain. Manometer: a U-shaped tube partly filled with a liquid, with one arm connected to a gas supply. The difference in liquid levels in the two arms measures the pressure of the gas above or below atmospheric pressure. It is used to measure gas pressures in laboratories and in gas supply lines.

D. Comprehensive Questions

6.3 Describe Pascal's Law. State its applications with examples.

Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the walls of the container. Applications: (1) hydraulic press and hydraulic lift: a small force on a small piston creates a pressure that acts on a large piston to lift a heavy load; (2) hydraulic brakes: pedal pressure is transmitted through brake fluid to all wheels equally; (3) hydraulic jacks used to lift vehicles for repair.

D. Comprehensive Questions

6.4 On what factors the pressure of a liquid in a container depend? How is it determined?

The pressure of a liquid at a point depends on three factors: the density of the liquid, the depth of the point below the free surface, and the acceleration due to gravity. It is given by P = rho g h, and it does not depend on the shape of the container or the total volume of liquid. It is determined either by calculation from

rho, g and h, or by measurement with a pressure gauge or a manometer connected to the container.

D. Comprehensive Questions

6.5 Explain that atmosphere exerts pressure. What are its applications? Give at least three examples.

The atmosphere is a thick layer of air with weight, and that weight presses on everything beneath it. Evidence: a drinking straw works because sucking removes air from the straw and atmospheric pressure pushes the drink up; a rubber suction cup sticks because air is squeezed out and atmospheric pressure holds it; and a mercury barometer column is supported entirely by atmospheric pressure. Applications: drinking through a straw, suction cups and vacuum lifters, the working of a siphon, and filling a fountain pen or a syringe.

E. Numerical Problems

6.1 A spring is stretched 20 mm by a load of 40 N. Calculate the value of spring constant. If an object causes an extension of 16 mm, what will be its weight? (Book's printed answers: 2 kN m^-1, 32 N)

Step 1: extension x = 20 mm = 0.020 m, force F = 40 N. Step 2: spring constant k = F/x = 40/0.020 = 2000 N m^-1 = 2 kN m^-1. Step 3: for extension 16 mm = 0.016 m, weight W = k x = 2000 x 0.016 = 32 N. Both match the book's printed answers.

E. Numerical Problems

6.2 The mass of 5 litres of milk is 4.5 kg. Find its density in SI units. (Book's printed answer: 0.9 x 10^3 kg m^-3)

Step 1: volume V = 5 litres = 5 x 10^-3 m^3. Step 2: density rho = mass/volume = 4.5 / (5 x 10^-3) = 900 kg m^-3 = 0.9 x 10^3 kg m^-3. This

matches the book's printed answer.

E. Numerical Problems

6.3 When a solid of mass 60 g is lowered into a measuring cylinder, the level of water rises from 40 cm^3 to 44 cm^3. Calculate the density of the solid. (Book's printed answer: 15 x 10^3 kg m^-3)

Step 1: volume of solid = rise in water level = 44 - 40 = 4 cm^3 = 4 x 10^-6 m^3. Step 2: mass = 60 g = 0.060 kg. Step 3: density = 0.060 / (4 x 10^-6) = 15,000 kg m^-3 = 15 x 10^3 kg m^-3. This matches the book's printed answer.

E. Numerical Problems

6.4 A brick measures 5 cm x 10 cm x 20 cm. If its mass is 5 kg, calculate the maximum and minimum pressure which the brick can exert on a horizontal surface. (Book's printed answers: 1 x 10^4 Pa, 25 x 10^2 Pa)

Step 1: weight of brick = 5 x 10 = 50 N. Step 2: smallest face area = 5 cm x 10 cm = 0.005 m^2; maximum pressure = 50/0.005 = 10,000 Pa = 1 x 10^4 Pa. Step 3: largest face area = 10 cm x 20 cm = 0.02 m^2; minimum pressure = 50/0.02 = 2500 Pa = 25 x 10^2 Pa. Both match the book's printed answers.

E. Numerical Problems

6.5 What will be the height of the column in barometer at sea level if mercury is replaced by water of density 1000 kg m^-3, where density of mercury is 13.6 x 10^3 kg m^-3? (Book's printed answer: 10.3 m)

Step 1: the same atmospheric pressure supports both columns, so rho_water x g x h_water = rho_mercury x g x h_mercury. Step 2: h_water = h_mercury x (rho_mercury / rho_water) = 0.76 x (13.6 x 10^3 / 1000) = 0.76 x 13.6 = 10.336 m, about 10.3 m. This matches the book's printed answer.

E. Numerical Problems

6.6 Suppose in the hydraulic brake system of a car, the force exerted normally on its piston of cross-sectional area of 5 cm^2 is 500 N. What will be the pressure transferred to the brake oil? What will be the force on the second piston of area of cross section 20 cm^2? (Book's printed answers: 1.0 x 10^6 N m^-2, 2000 N)

Step 1: area A1 = 5 cm^2 = 5 x 10^-4 m^2. Step 2: pressure P = F/A = 500 / (5 x 10^-4) = 1.0 x 10^6 N m^-2; by Pascal's law this pressure is transferred unchanged to the brake oil. Step 3: area A2 = 20 cm^2 = 20 x 10^-4 m^2. Step 4: force on second piston F2 = P x A2 = 1.0 x 10^6 x 20 x 10^-4 = 2000 N. Both match the book's printed answers.

E. Numerical Problems

6.7 The area of cross-section of the small and large pistons of a hydraulic press is respectively 10 cm^2 and 100 cm^2. What force should be exerted on the small piston in order to lift a car of weight 4000 N? (Book's printed answer: 400 N)

Step 1: by Pascal's law, pressure is the same on both pistons: F1/A1 = F2/A2. Step 2: F1 = F2 x (A1/A2) = 4000 x (10/100) = 400 N. This matches the book's printed answer.

E. Numerical Problems

6.8 If the pressure in a hydraulic press is increased by an additional 10 N cm^-2, how much extra load will the output platform support if its cross-sectional area is 50 cm^2? (Book's printed answer: 500 N)

Step 1: extra force = increase in pressure x area. Step 2: extra force = 10 N cm^-2 x 50 cm^2 = 500 N. This matches the book's printed answer.

E. Numerical Problems

6.9 The force exerted normally on the hydraulic brake system of a car, with its piston of cross sectional area 5 cm^2 is 500 N. What will be the: (a) pressure transferred to the brake oil? (b) force on the brake piston of area of cross section 20 cm^2? (Book's printed answers: (a) 1.0 x 10^6 N m^-2, (b) 2000 N)

This problem repeats problem 6.6 as printed in the book, and is solved the same way. (a) Pressure P = F/A = 500 / (5 x 10^-4) = 1.0 x 10^6 N m^-2, transferred unchanged to the brake oil. (b) Force on the brake piston = P x A = 1.0 x 10^6 x 20 x 10^-4 = 2000 N. Both match the book's printed answers.

Chapter 7: Thermal Properties of Matter

A. Multiple Choice Questions

7.1 How do the molecules in a solid behave?

(a) Move randomly

(b) Vibrate about their mean positions ✓

(c) Rotate and vibrate randomly at their own positions

(d) Move in a straight line from hot to cold ends.

Correct answer: Vibrate about their mean positions

A. Multiple Choice Questions

7.2 What type of motion is of the molecules in a gas?

(a) Linear motion

(b) Random motion ✓

(c) Vibratory motion

(d) Rotatory motion

Correct answer: Random motion

A. Multiple Choice Questions

7.3 Temperature of a substance is:

(a) the total amount of heat contained in it

(b) the total number of molecules in it

(c) degree of hotness or coldness ✓

(d) dependent upon the intermolecular distance

Correct answer: degree of hotness or coldness

A. Multiple Choice Questions

7.4 Heat is the:

(a) total kinetic energy of the molecules

(b) the internal energy

(c) work done by the molecules

(d) the energy in transit ✓

Correct answer: the energy in transit

A. Multiple Choice Questions

7.5 In Kelvin scale, the temperature corresponding to melting point of ice is:

(a) zero

(b) 32

(c) -273

(d) +273 ✓

Correct answer: +273

A. Multiple Choice Questions

7.6 The temperature which has the same value on Celsius and Fahrenheit scale is:

(a) -40 ✓

(b) +40

(c) +45

(d) -45

Correct answer: -40

Working for Question 7.6

Working: the temperature common to Celsius and Fahrenheit

Let the common value be C = F = T. The conversion is F = (9/5) C + 32. Putting F = C = T gives T = (9/5) T + 32, so T - (9/5) T = 32, giving -(4/5) T = 32 and T = -40. So -40 degrees is the same on both scales, option (a).

A. Multiple Choice Questions

7.7 Which one is a better choice for a liquid-in-glass thermometer?

(a) Is colourless

(b) Is a bad conductor

(c) Expand linearly ✓

(d) Wets glass

Correct answer: Expand linearly

A. Multiple Choice Questions

7.8 One disadvantage of using alcohol in a liquid-in-glass thermometer:

(a) it has large expansivity

(b) it has low freezing point (-112 degrees C)

(c) it wets the glass tube ✓

(d) its expansion is linear

Correct answer: it wets the glass tube

A. Multiple Choice Questions

7.9 Water is not used as a thermometric liquid mainly due to:

(a) colourless

(b) a bad conductor of heat

(c) non-linear expansion ✓

(d) a low boiling point (100 degrees C)

Correct answer: non-linear expansion

A. Multiple Choice Questions

7.10 A thermometer has a narrow capillary tube so that it:

(a) quickly responds to temperature changes

(b) can read the maximum temperature

(c) gives a large change for a given temperature rise

(d) can measure a large range of temperature

Correct answer: gives a large change for a given temperature rise

B. Short Answer Questions

7.1 Why solids have a fixed volume and shape according to particle theory of matter?

In a solid the particles are held close together by strong forces and can only vibrate about fixed mean positions; they cannot move past one another. The fixed arrangement of particles therefore keeps both the shape and the volume fixed.

B. Short Answer Questions

7.2 What are the reasons that gases have neither a fixed volume nor a fixed shape?

Gas particles are far apart, the forces between them are negligible, and they move freely and randomly at high speeds in all directions. Nothing holds them in fixed positions, so a gas spreads out to fill the whole container, taking the container's shape and volume.

B. Short Answer Questions

7.3 Compare the spacing of molecules in the solid, liquid and gaseous state.

In a solid the molecules are packed very closely together in a regular arrangement. In a liquid they are still close together but less regularly arranged and free to slide past one another. In a gas they are very far apart compared with their own size, with large empty spaces between them.

B. Short Answer Questions

7.4 What is the effect of raising the temperature of a liquid?

Raising the temperature increases the average kinetic energy of the liquid's molecules, so they vibrate and move faster, push slightly further apart, and the liquid expands. If heating continues, more molecules gain enough energy to escape from the surface, so evaporation increases, and at the boiling point the liquid changes into vapour.

B. Short Answer Questions

7.5 What is meant by temperature of a body?

The temperature of a body is the measure of the average kinetic energy of its molecules, and it tells how hot or cold the body is. Two bodies at the same temperature are in thermal equilibrium, and heat flows from the body at higher temperature to the body at lower temperature.

B. Short Answer Questions

7.6 Define heat as 'energy in transit'.

Heat is energy that flows from one body to another because of a temperature difference. It is called energy in transit because it exists only while it is flowing: a body does not contain heat, it contains internal energy, and heat is the name for internal energy on the move from a hotter body to a colder one.

B. Short Answer Questions

7.7 What is meant by thermometric property of a substance? Enlist some thermometric properties.

A thermometric property is a physical property of a substance that changes regularly and measurably with temperature, so it can be used to measure temperature. Examples: the length of a mercury column in a glass tube, the volume of a gas at constant pressure, the pressure of a gas at constant volume,

the electrical resistance of a metal wire, and the voltage produced by a thermocouple.

B. Short Answer Questions

7.8 State the main scales used for the measurement of temperature.

The three main scales are: the Celsius scale (ice point 0 degrees C, steam point 100 degrees C); the Fahrenheit scale (ice point 32 degrees F, steam point 212 degrees F); and the Kelvin scale, the SI scale (ice point 273 K, steam point 373 K). The Kelvin scale starts at absolute zero, 0 K = -273 degrees C.

B. Short Answer Questions

7.9 What is meant by sensitivity of a thermometer?

Sensitivity is how large a change the thermometer shows for a small change in temperature. A sensitive thermometer gives a large, easily read movement of its indicator, such as a long movement of the mercury thread, for even a small rise in temperature.

B. Short Answer Questions

7.10 What do you mean by the linearity of a thermometer?

Linearity means that equal rises in temperature produce equal changes in the thermometric property, so the scale divisions are equally spaced. A linear thermometer can be calibrated with just two fixed points and read accurately anywhere between them; a non-linear thermometer needs many calibration points.

B. Short Answer Questions

7.11 What makes the scale reading of a thermometer accurate?

An accurate reading needs a thermometer that is properly calibrated against

fixed points, has a linear and sensitive thermometric property, and is given enough time to reach thermal equilibrium with the body being measured, with good thermal contact and no heat leaking away during the reading.

B. Short Answer Questions

7.12 What does determine the direction of heat flow?

The temperature difference determines the direction of heat flow: heat always flows from the body at higher temperature to the body at lower temperature, never the other way by itself. The flow continues until both bodies reach the same temperature.

B. Short Answer Questions

7.13 Distinguish between the heat and internal energy.

Internal energy is the total energy stored in a body, the sum of the kinetic and potential energies of all its molecules; a body contains internal energy at all times. Heat is not stored in a body at all: it is energy in transit, flowing from a hotter body to a colder one because of a temperature difference. When heat flows into a body, the body's internal energy increases; when heat flows out, it decreases.

B. Short Answer Questions

7.14 When you touch a cold surface, does cold travel from the surface to your hand or does energy travel from your hand to cold surface?

Energy travels from your hand to the cold surface. Cold is not a thing that flows; heat is energy in transit, and it always flows from the warmer body (your hand) to the colder body (the surface). The loss of energy from your hand is what you feel as coldness.

B. Short Answer Questions

7.15 Can you feel your fever by touching your own forehead? Explain.

No, not reliably. Your hand and your forehead are both parts of your own body at nearly the same temperature, so almost no heat flows between them and your hand feels no clear difference. A thermometer, or another person's cooler hand, can detect the fever because a real temperature difference exists for heat to flow across.

C. Constructed Response Questions

7.1 Is kinetic molecular theory of matter applicable to the plasma state of matter? Describe briefly.

Yes. Plasma is a fourth state of matter consisting of ionised gas: free electrons and positive ions moving randomly at high speeds. The kinetic molecular theory still applies, because it describes matter as particles in random motion whose average kinetic energy fixes the temperature; in plasma the particles are simply charged, and their collisions and electric forces are added to the picture. The Sun and the stars are matter in the plasma state, described by the same kinetic ideas.

C. Constructed Response Questions

7.2 Why is mercury usually preferred to alcohol as a thermometric liquid?

Mercury is preferred because it does not wet the glass tube, so the thread moves cleanly; it is opaque and shiny, so it is easy to see; it expands almost linearly with temperature; and it has a wide liquid range, from -39 degrees C to 357 degrees C, covering most everyday and laboratory temperatures. Alcohol wets the glass, is colourless and must be dyed, and boils at a low temperature, so it is used only for very low temperatures where mercury would freeze.

C. Constructed Response Questions

7.3 Why is water not suitable for use in thermometers? Without calculations, guess what is equivalent temperature of 373 K on Celsius and Fahrenheit scales?

Water is unsuitable because its expansion is not linear: it expands unevenly, and between 0 degrees C and 4 degrees C it actually contracts as it warms, so the same thermometer reading could mean two different temperatures. It also wets the glass, is colourless and hard to see, and has a narrow liquid range for a thermometer. The equivalent of 373 K is 100 degrees C on the Celsius scale (373 - 273) and 212 degrees F on the Fahrenheit scale, the steam point of water.

C. Constructed Response Questions

7.4 Mention two ways in which the design of a liquid-in-glass thermometer may be altered to increase its sensitivity.

Two ways: (1) make the capillary tube narrower, so the same expansion of liquid in the bulb pushes the thread much further along the tube; (2) make the bulb larger, so a given temperature rise expands a larger volume of liquid and drives the thread further. Using a liquid with a larger expansivity also increases sensitivity.

C. Constructed Response Questions

7.5 One litre of water is heated by a stove and its temperature rises by 2 degrees C. If two litres of water is heated on the same stove for the same time, what will be then rise in temperature?

The rise will be 1 degree C. The stove supplies the same amount of heat in the same time, but the heat is now shared by twice the mass of water. Since temperature rise = heat supplied / (mass x specific heat capacity), doubling the mass halves the rise: 2 degrees C / 2 = 1 degree C.

C. Constructed Response Questions

7.6 Why are there no negative numbers on the Kelvin scale?

The Kelvin scale starts at absolute zero, 0 K, the temperature at which the molecules of a substance have their minimum possible kinetic energy and cannot lose any more. No body can be colder than absolute zero, because kinetic energy cannot be negative, so no temperature below 0 K exists and the scale needs no negative numbers. 0 K equals -273 degrees C.

C. Constructed Response Questions

7.7 Comment on the statement. 'A thermometer measures its own temperature.'

The statement is true in an important sense. A thermometer does not directly read the temperature of the body it touches; it reads the temperature it itself reaches after exchanging heat with that body. Only when the thermometer has been in good contact long enough to reach thermal equilibrium with the body does its own temperature equal the body's temperature, and only then is its reading the body's temperature.

C. Constructed Response Questions

7.8 There are various objects made of cotton, wood, plastic, metals, etc. In a winter night, compare their temperatures with the air temperature by touching them with your hand.

After a long winter night, all the objects have reached thermal equilibrium with the air, so they are all actually at the same temperature as the air. They feel different only because metals conduct heat away from your warm hand quickly and feel cold, while cotton, wood and plastic conduct heat slowly and feel warmer. Touch therefore judges how fast heat leaves your hand, not the true temperature; a thermometer would show all of them at the air temperature.

C. Constructed Response Questions

7.9 Which is greater: an increase in temperature 1 degree C or one 1 degree F?

An increase of 1 degree C is greater. One Celsius degree equals 1.8 Fahrenheit degrees, because the Celsius scale has 100 divisions between the ice and steam points while the Fahrenheit scale has 180 divisions over the same interval. So a rise of 1 degree C is the same as a rise of 1.8 degrees F, which is larger than a rise of 1 degree F.

C. Constructed Response Questions

7.10 Why would not you expect all the molecules in a gas to have the same speed?

Gas molecules constantly collide with one another and with the walls, and in every collision energy is shared unequally: some molecules speed up and others slow down. The result is a spread of speeds, described by the Maxwell distribution, with most molecules near the average speed and a few much faster or much slower. Only the average speed is fixed by the temperature; individual speeds keep changing.

C. Constructed Response Questions

7.11 Does it make sense to talk about the temperature of a vacuum?

No, not in the usual sense. Temperature measures the average kinetic energy of particles, and a perfect vacuum contains no particles at all, so it has no average kinetic energy to measure. We can speak of the temperature of the walls of a vacuum vessel, or of radiation passing through it, but the empty space itself has no temperature of its own.

C. Constructed Response Questions

7.12 Comment on the statement: 'A hot body does not contain heat'.

The statement is correct. Heat is energy in transit, flowing because of a temperature difference; it is not something stored inside a body. A hot body contains internal energy, the kinetic and potential energy of its molecules, which is large because its temperature is high. When it cools, it loses internal energy by giving out heat, but at no moment does it contain heat as a stored quantity.

C. Constructed Response Questions

7.13 Discuss whether the Sun is matter.

Yes, the Sun is matter. It is made of particles, mostly hydrogen and helium nuclei and free electrons, in the plasma state: a very hot, ionised gas. Plasma is a genuine state of matter, like solid, liquid and gas, with mass and volume, and the Sun's enormous mass holds it together by gravity. Being plasma does not stop it being matter; it is simply matter at extremely high temperature.

D. Comprehensive Questions

7.1 Describe the main points of particle theory of matter which differentiate solids, liquids and gases.

Particle theory says all matter is made of tiny particles in constant motion, with forces of attraction between them. In solids the particles are close together, strongly attracted, and vibrate about fixed positions, giving a fixed shape and volume. In liquids the particles are still close but the forces are weaker, so they can slide past one another: the volume stays fixed but the shape changes to fit the container. In gases the particles are far apart, the forces are negligible, and they move freely and randomly: neither shape nor volume is fixed, and the gas fills its container.

D. Comprehensive Questions

7.2 What is temperature? How is it measured? Describe briefly the construction

of a mercury-in-glass thermometer.

Temperature is the measure of the average kinetic energy of the molecules of a body, telling how hot or cold it is. It is measured with a thermometer using a thermometric property that changes with temperature. A mercury-in-glass thermometer has a thin-walled glass bulb containing mercury, joined to a very narrow, uniform capillary tube sealed at the top, with a scale marked along the stem. When the bulb warms, the mercury expands and rises along the capillary; the position of the thread against the scale gives the temperature. The bulb is thin so heat reaches the mercury quickly, and the capillary is narrow so a small expansion gives a large, readable movement.

D. Comprehensive Questions

7.3 Compare the three scales used for measuring temperature.

The Celsius scale fixes the ice point at 0 degrees C and the steam point at 100 degrees C, with 100 equal divisions between them; it is the everyday scientific scale. The Fahrenheit scale fixes the ice point at 32 degrees F and the steam point at 212 degrees F, with 180 divisions between them; it is used mainly in a few countries for weather and daily life. The Kelvin scale is the SI scale: it starts at absolute zero, 0 K = -273 degrees C, with the ice point at 273 K and the steam point at 373 K, and its divisions are the same size as Celsius degrees. Conversions: K = C + 273, and F = (9/5) C + 32.

D. Comprehensive Questions

7.4 What is meant by sensitive, range and linearity of thermometers? Explain with examples.

Sensitivity is the size of the thermometer's response to a small temperature change: a clinical thermometer is highly sensitive because its narrow capillary shows a large thread movement for a small fever change. Range is the interval

of temperatures the thermometer can measure: a laboratory mercury thermometer ranges from about -39 degrees C to 357 degrees C, while a clinical thermometer covers only about 35 degrees C to 42 degrees C. Linearity means equal temperature rises give equal scale movements: mercury expands almost linearly, so its scale divisions are equal, whereas a thermometer using a property that changes unevenly would need unequal divisions and careful calibration.

E. Numerical Problems

7.1 The temperature of a normal human body on Fahrenheit scale is 98.6 degrees F. Convert it into Celsius scale and Kelvin scale. (Book's printed answers: 37 degrees C, 310 K)

Step 1: Celsius C = (F - 32) x 5/9 = (98.6 - 32) x 5/9 = 66.6 x 5/9 = 37 degrees C. Step 2: Kelvin K = C + 273 = 37 + 273 = 310 K. Both match the book's printed answers.

E. Numerical Problems

7.2 At what temperature Celsius and Fahrenheit thermometer reading would be the same? (Book's printed answer: -40 degrees)

Step 1: let the common value be T, so C = F = T. Step 2: F = (9/5) C + 32 becomes T = (9/5) T + 32. Step 3: T - (9/5) T = 32, so -(4/5) T = 32, giving T = -40 degrees. This matches the book's printed answer.

E. Numerical Problems

7.3 Convert 5 degrees F to Celsius and Kelvin scale. (Book's printed answers: -15 degrees C, 258 K)

Step 1: C = (5 - 32) x 5/9 = (-27) x 5/9 = -15 degrees C. Step 2: K = C + 273 = -15 + 273 = 258 K. Both match the book's printed answers.

E. Numerical Problems

7.4 What is equivalent temperature of 25 degrees C on Fahrenheit and Kelvin scales? (Book's printed answers: 77 degrees F, 298 K)

Step 1: F = (9/5) x 25 + 32 = 45 + 32 = 77 degrees F. Step 2: K = 25 + 273 = 298 K. Both match the book's printed answers.

E. Numerical Problems

7.5 The ice and steam points on an ungraduated thermometer are found to be 192 mm apart. What temperature will be on Celsius scale if the length of mercury thread is at 67.2 mm above the ice point mark? (Book's printed answer: 35 degrees C)

Step 1: the interval between ice point and steam point is 192 mm for 100 degrees C. Step 2: temperature above the ice point = (length above ice point / total length) x 100 = (67.2/192) x 100. Step 3: 67.2/192 = 0.35, so temperature = 0.35 x 100 = 35 degrees C. This matches the book's printed answer.

E. Numerical Problems

7.6 The length between the fixed point of liquid-in-glass thermometer is 20 cm. If the mercury level is 4.5 cm above the lower mark, what is the temperature on the Fahrenheit scale? (Book's printed answer: 72.5 degrees F)

Step 1: the fixed points are the ice point (32 degrees F) and the steam point (212 degrees F), a range of 180 Fahrenheit degrees over 20 cm. Step 2: fraction above the lower mark = 4.5/20 = 0.225. Step 3: temperature = 32 + 0.225 x 180 = 32 + 40.5 = 72.5 degrees F. This matches the book's printed answer.

Chapter 8: Magnetism

A. Multiple Choice Questions

8.1 Which one of the following is not a magnetic material?

(a) Cobalt

(b) Iron

(c) Aluminium ✓

(d) Nickel

Correct answer: Aluminium

A. Multiple Choice Questions

8.2 Magnetic lines of force:

(a) are always directed in a straight line

(b) cross one another

(c) enter into the north pole

(d) enter into the south pole ✓

Correct answer: enter into the south pole

Working for Question 8.2

Working: direction of magnetic lines of force

Outside a magnet, magnetic lines of force leave the north pole and enter the south pole, then continue inside the magnet from south back to north, forming closed loops. They are curved, not always straight, and they never cross one another. So option (d) is correct.

A. Multiple Choice Questions

8.3 Permanent magnets cannot be made by:

(a) soft iron ✓

(b) steel

(c) neodymium

(d) alnico

Correct answer: soft iron

Working for Question 8.3

Working: why can soft iron not make a permanent magnet?

Soft iron is easily magnetised but loses its magnetism just as easily when the magnetising force is removed, so it can only make temporary magnets. Steel, neodymium and alnico retain their magnetism once magnetised, so they are used for permanent magnets. Hence option (a) is correct.

A. Multiple Choice Questions

8.4 Permanent magnets are used in:

(a) circuit breaker

(b) loudspeaker ✓

(c) electric crane

(d) magnetic recording

Correct answer: loudspeaker

A. Multiple Choice Questions

8.5 A common method used to magnetise a material is:

(a) stroking ✓

(b) hitting

(c) heating

(d) placing inside a solenoid having A.C

Correct answer: stroking

A. Multiple Choice Questions

8.6 A magnetic compass is placed around a bar magnet at four points as shown in figure below. Which diagram would indicate the correct directions of the field?

(a) Diagram (a): top arrow right, bottom arrow left, left arrow left, right arrow right

(b) Diagram (b): top arrow left, bottom arrow right, left arrow left, right arrow left

(c) Diagram (c): top arrow right, bottom arrow right, left arrow left, right arrow left ✓

(d) Diagram (d): top arrow left, bottom arrow left, left arrow left, right arrow right

Correct answer: Diagram (c): top arrow right, bottom arrow right, left arrow left, right arrow left

Working for Question 8.6

Working: compass directions around a bar magnet

The field outside a magnet runs from its north pole to its south pole. To the left of the north pole the field points away from the magnet, to the left. To the right of the south pole the field points into the magnet, also to the left. Above the magnet the field lines travel from north to south, to the right, and below the magnet they also travel from the north side towards the south side, to the right. So the correct pattern is top right, bottom right, left left, right left, which is diagram (c).

A. Multiple Choice Questions

8.7 A steel rod is magnetised by double touch stroking method. Which one would be the correct polarity of the AB magnet?

(a) A = N, B = S

(b) A = S, B = N ✓

(c) A = N with two S poles in the middle, B = N

(d) A = S with two N poles in the middle, B = S

Correct answer: A = S, B = N

Working for Question 8.7

Working: polarity from double-touch stroking

In the figure the two stroking magnets start at the centre of the bar and move outward: the left magnet carries its S pole in contact with the bar towards end A, and the right magnet carries its N pole in contact towards end B. In stroking, the end where a magnet finishes its stroke takes the same polarity as the stroking pole. So end A becomes S and end B becomes N, giving the AB magnet with A = S and B = N, option (b).

A. Multiple Choice Questions

8.8 The best material to protect a device from external magnetic field is:

(a) wood

(b) plastic

(c) steel

(d) soft iron ✓

Correct answer: soft iron

Working for Question 8.8

Working: magnetic shielding

Soft iron has a very high magnetic permeability, so magnetic flux prefers to travel through it rather than through air. A soft-iron casing therefore carries the external field around the device inside it instead of letting it pass through, shielding the device. Wood and plastic have no such effect, and steel retains magnetism, so soft iron, option (d), is the best shield.

B. Short Answer Questions

8.1 What are temporary and permanent magnets?

A temporary magnet is magnetised only while a magnetising force acts on it and loses its magnetism soon after the force is removed; soft iron is the usual material. A permanent magnet keeps its magnetism for a long time after being magnetised; steel, alnico and neodymium are the usual materials.

B. Short Answer Questions

8.2 Define magnetic field of a magnet.

The magnetic field of a magnet is the region around it in which a magnetic force acts on another magnet or on a magnetic material. It is mapped by magnetic lines of force, which show both the direction and the relative strength of the field at each point.

B. Short Answer Questions

8.3 What are magnetic lines of force?

Magnetic lines of force are imaginary lines used to picture a magnetic field. A compass needle placed on such a line points along it. Their properties are: they leave the north pole and enter the south pole outside the magnet; they never cross one another; they are crowded together where the field is strong and spread apart where it is weak; and they always form closed loops through the inside of the magnet.

B. Short Answer Questions

8.4 Name some uses of permanent magnets and electromagnets.

Permanent magnets are used in compasses, loudspeakers, electric motors and generators, refrigerator door seals, and magnetic toys. Electromagnets, whose magnetism can be switched on and off, are used in electric cranes for lifting scrap iron, electric bells, relays and circuit breakers, magnetic separators, and

the cores of transformers and motors.

B. Short Answer Questions

8.5 Which type of magnetic field is formed by a current-carrying long coil?

A current-carrying long coil, or solenoid, produces a magnetic field like that of a bar magnet: the field inside the coil is strong and nearly uniform, with parallel lines running from one end to the other, and the field outside loops back from the north-seeking end to the south-seeking end. The polarity of the ends is fixed by the direction of the current.

C. Constructed Response Questions

8.1 Two bar magnets are stored in a wooden box. Label the poles of the magnets and identify P and Q objects.

In the figure the left magnet is shown with its upper half red and its lower half blue, and the right magnet with its upper half blue and its lower half red. Taking red as north and blue as south, the left magnet has its N pole at the top and its S pole at the bottom, while the right magnet has its S pole at the top and its N pole at the bottom, so unlike poles face each other across the top and the bottom. P and Q, the bars laid across the tops and bottoms of the two magnets, are soft-iron keepers: they join the opposite poles and close the magnetic circuit, which stops the magnets from losing their magnetism during storage.

C. Constructed Response Questions

8.2 A steel bar has to be magnetised by placing it inside a solenoid such that end A of a bar becomes N-pole and end B becomes S-pole. Draw circuit diagram of solenoid showing steel bar inside it.

Circuit diagram to draw: a solenoid drawn as a horizontal coil with the steel bar AB lying inside it along its axis, a battery connected to the ends of the coil

through a switch, and the current direction marked. To make end A (the left end) a north pole, the current viewed from the A end must flow anticlockwise, by the right-hand grip rule: curl the fingers of the right hand in the direction of the current and the thumb points to the north pole. So wire the battery so that current enters the coil at the B end side and circulates anticlockwise as seen from A; label end A as N, end B as S, and mark the battery terminals and the current arrows on the coil.

C. Constructed Response Questions

8.3 Two bar magnets are lying as shown in the figure. A compass is placed at the middle of the gap. Its needle settles in the north-south direction. Label N and S poles of the magnets. Justify your answer by drawing fields lines.

The two facing ends must be like poles. Justification: if the facing ends were unlike poles, the field in the gap would point straight across the gap from the north pole to the south pole, and the compass needle would settle pointing across the gap, east-west. Instead the needle settles north-south, the direction of the Earth's own field, which means the two magnets' fields cancel each other at the middle of the gap. That cancellation happens only when the facing ends are like poles, both north or both south, pushing equal fields against each other. Field lines to draw: lines leaving each facing north pole (or entering each facing south pole) and curving away, meeting and cancelling at the centre, with the compass needle drawn north-south at the centre. Note: the needle alone proves the facing poles are alike, but it cannot by itself tell whether they are both north or both south.

C. Constructed Response Questions

8.4 Electric current or motion of electrons produce magnetic field. Is the reverse process true, that is the magnetic field gives rise to electric current? If yes, give

an example and describe it briefly.

Yes, the reverse process is true and is called electromagnetic induction. When the magnetic field through a coil changes, a current is induced in the coil. Example: a generator or dynamo. A coil rotated between the poles of a magnet experiences a continuously changing magnetic field, so a current is induced in the coil and can light a bulb or charge a battery. Moving a bar magnet in and out of a coil has the same effect: the galvanometer needle deflects while the magnet moves and returns to zero when it stops, showing that only a changing field induces a current.

C. Constructed Response Questions

8.5 Four similar solenoids are placed in a circle as shown in the figure. The magnitude of current in all of them should be the same. Show by diagram, the direction of current in each solenoid such that when current in anyone solenoid is switched OFF, the net magnetic field at the centre O is directed towards that solenoid. Explain your answer.

Diagram to draw: four solenoids A, B, C and D placed at the left, bottom, right and top of a circle around centre O, each with its axis pointing at O, and each wound so that its inner end facing O is a north pole; mark the current direction in each coil accordingly (anticlockwise as seen from O for each inner end).

Explanation: each solenoid's field at O points radially outward, away from its own north pole, and the four equal outward fields cancel one another, so the net field at O is zero with all solenoids on. When one solenoid, say D at the top, is switched off, its outward (downward) contribution disappears, and the remaining three fields no longer cancel: their net field points in the opposite direction to the missing contribution, that is upward, towards the switched-off solenoid D. The same argument holds for whichever solenoid is switched off.

D. Comprehensive Questions

8.1 How can you identify whether an object is a magnet or a magnetic material?

Suspend the object freely: a magnet always settles pointing roughly north-south, while an unmagnetised magnetic material has no preferred direction. Bring a known magnet near it: a magnet is repelled by a like pole of the known magnet, while a mere magnetic material is only ever attracted, never repelled. Repulsion is therefore the sure test: attraction alone cannot distinguish a magnet from a magnetic material, because a magnet attracts both magnets and magnetic materials.

D. Comprehensive Questions

8.2 Describe the strength of a magnetic field in terms of magnetic lines of force. Explain it by drawing a few diagrams for the fields as examples.

The strength of a magnetic field at a point is shown by how crowded the magnetic lines of force are there: closely spaced lines mean a strong field, widely spaced lines mean a weak field, and parallel, equally spaced lines mean a uniform field. Diagrams to draw: (1) a bar magnet with lines crowded near the poles and spreading out further away, showing the field is strongest at the poles; (2) two like poles facing each other, with lines bending away from the gap, showing repulsion and zero field at the middle; (3) a uniform field drawn as parallel, equally spaced straight lines, as inside a long solenoid. In each diagram label where the field is strong and where it is weak.

D. Comprehensive Questions

8.3 A magnet attracts only a magnet. Explain the statement.

Taken literally the statement is not complete, and it should be read as: a magnet attracts only magnetic materials. A magnet attracts iron, nickel, cobalt and other ferromagnetic materials, whether or not they are already magnets, because it

first induces magnetism in them and then attracts them. It does not attract non-magnetic materials such as wood, plastic, aluminium or copper. The statement is therefore true only in the sense that attraction is limited to magnetic substances; a magnet does not attract everything, and repulsion, not attraction, is the test that proves another body is itself a magnet.

Note on Chapter 8

Note: Chapter 8 has no Numerical Problems section

The Chapter 8 exercise in the book (book pp. 128-130) ends with section D, Comprehensive Questions 8.1-8.3, and the Glossary begins on book p. 131. No section E, Numerical Problems, is printed for this chapter, so none is included here and nothing has been invented to fill the gap.